

# The effects of implicit causality, topicality, and controller animacy on temporal adjunct control

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## Abstract

Non-finite temporal adjuncts often only allow control by the matrix subject, but recent research has shown that non-subject control is also possible. Previous research has found that object control is more likely if the object's referent is strongly topical. Other research predicts that it should also be possible if the object is logophoric. In three reference choice and acceptability rating experiments, this study demonstrates that object control of temporal adjuncts in English is more widely available than previously thought in sentences such as *Alyssa noticed Sergio after sitting in a front-row seat*. Object control interpretations are more likely with a supporting context, which can consist of topicality or logophoricity of the object, as well as with more generic referential biases like implicit causality. These results indicate that object control of temporal adjuncts is grammatical in English, even if subject control is often preferred. Exploratory analyses indicate that this cannot be due to structural ambiguity in the adjunct's attachment height and is unlikely to be a grammatical illusion. Instead, the results support theories arguing that non-subject controlled PRO in temporal adjuncts is simply a null pronoun *pro* whose reference is determined by several competing constraints.

## 1. Introduction

Temporal adjunct control involves a referential dependency between the null subject in a temporal adjunct clause (often represented as “PRO”) and its understood antecedent, as in the examples in (1).

- (1) (a) Hermione<sub>i</sub> confused Ron [after PRO<sub>i</sub> using the time turner].  
(b) Anna<sub>i</sub> spoke with Jeremy [while PRO<sub>i</sub> cooking dinner].

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Sentence-final adjunct control structures were historically treated as an Obligatory Control (OC) dependency in which PRO was required by the syntax to co-refer with the subject of the next highest clause (Mohanalan 1983, Clark 1990, Hornstein 1999, Pires 2007, McFadden & Sundaresan 2016). However, other research has shown that adjunct control structures also permit Non-Obligatory Control (NOC), in which PRO's referent is understood as someone not represented in the sentence, as illustrated in (2). And contrary to previous claims (McFadden & Sundaresan 2016, Fischer & Høyem 2022), NOC is not restricted to contexts in which OC is blocked (Landau 2017, 2021; Green 2019a,b). For example, (3) allows PRO either to be controlled by the subject or to refer to some sentence-external perspective holder.

- (2) (a) That oasis was a vision [after PRO dragging ourselves through the desert all day]. (Landau 2017: 7)
- (b) Mushrooms aren't too bad [after PRO being forced to eat celery]. (Green 2019b: 139)
- (3) The pool<sub>i</sub> was the perfect temperature [after PRO<sub>i/j</sub> being in the hot sun all day]. (Green 2019b: 140)

Control by the matrix object has also been claimed to be possible, as seen in (4). However, even when object control is available, it can be degraded, especially compared to subject control (Green 2018: 73, footnote 17). In fact, Gerard (2021) has argued that object control in temporal adjuncts is not grammatical in English, but at least in children result from an over-taxed parsing system.

- (4) (a) The security guard stopped the woman<sub>i</sub> [before PRO<sub>i</sub> boarding the plane]. (Green 2018: 73).
- (b) Woman<sub>i</sub>'s family beats abusive husband<sub>j</sub> with sticks [after PRO<sub>j</sub> leaving her<sub>i</sub> with black eye]. (Paz 2019)

The current study uses three acceptability judgment experiments to investigate whether object control is in fact available for native English speakers, as well as what constraints there are on its availability.

### *1.1. Previous research on object control in temporal adjuncts*

Janke & Bailey (2017) examined the effect of topicality on the interpretation of various types of control structures, including temporal adjunct control. The relevant manipulation for the current study is illustrated in (5). Participants were given a temporal adjunct control structure in one of three conditions: no prior context (5a), or prior context either weakly (5b) or strongly (5c) establishing the main clause object as a topic. After reading the sentence, participants were asked who performed the action in the adjunct clause (e.g., they were asked who fed the owl: Harry or Luna).

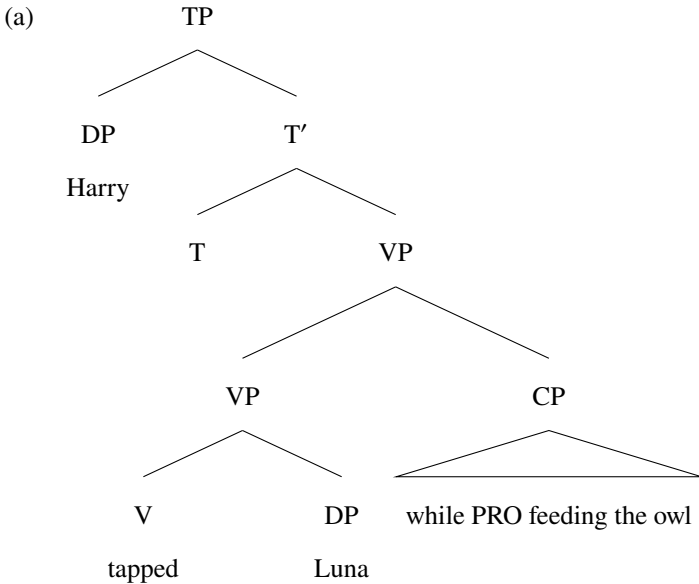
- (5) (a) Harry tapped Luna while feeding the owl.  
 (b) I'm going to tell you something about Luna. Harry tapped Luna while feeding the owl.  
 (c) Luna is looking after the birds. Luna takes out the food. Harry tapped Luna while feeding the owl.

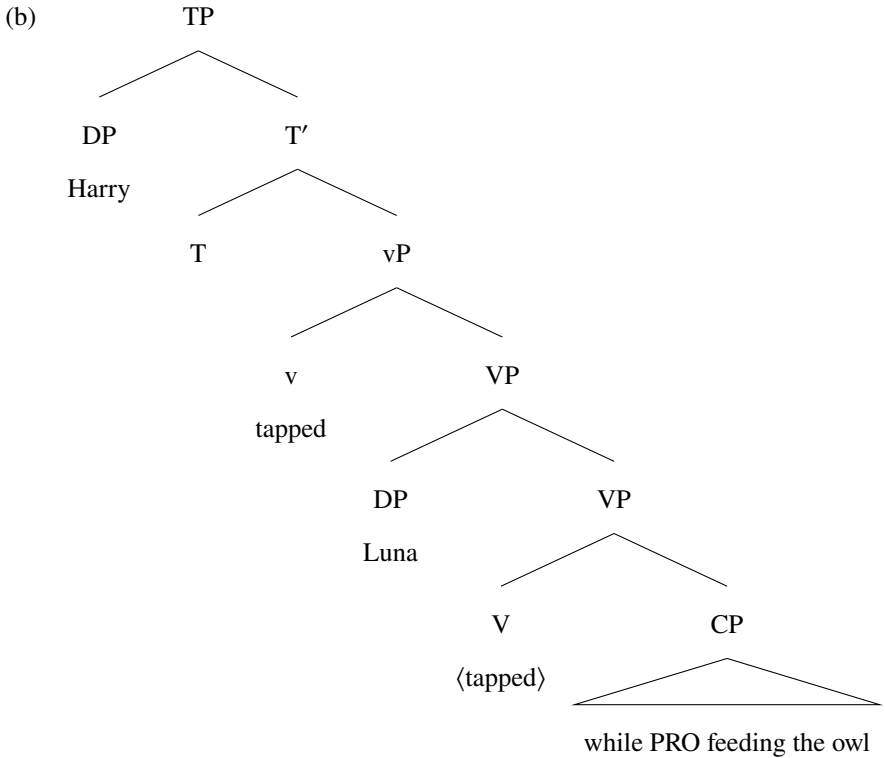
The results indicated that participants chose the object as PRO's antecedent (e.g., said that Luna fed the owl) in only 4% of trials without any prior context, in 11% of trials with the object as a weakly-established topic, and 51% of trials with the object as a strongly-established topic. This provided empirical evidence that although subject control of temporal adjuncts is preferred, topicality biases can reduce that preference, allowing object control to occur. However, one limitation of Janke & Bailey's results is that participants were given only a forced-choice reference resolution task. This gave evidence on whether they thought subject control or object control was more likely, but not how acceptable that reference resolution actually was. In the strong topic context in (5c), for example, some participants may have thought that Luna was more likely to be the one

feeding the owl, resulting in 51% object control responses, but that the sentence was still an unnatural or ungrammatical way to express that meaning.

Assuming that object control was available and acceptable, Janke & Bailey explain their findings by proposing a structural ambiguity between the subject and object control readings, as illustrated in (6). They make the fairly standard assumption (see also [Hornstein 1999](#), [Green 2019a](#), [Landau 2021](#), [Fischer & Høyem 2022](#)) that subject-controlled temporal adjuncts adjoin at a level outside the main VP containing the verb and its direct object, as illustrated in (6a). For object control, they follow [Larson \(2004\)](#) in assuming that the adjunct adjoins lower, within the VP, as in (6b), with the verb raising from V to v. They argue that subject control is preferred due to its simpler structure, including a lack of V-to-v movement.

(6) Temporal adjunct control structures (adapted from [Janke & Bailey 2017](#): 556–557)





However, this structural ambiguity account of object control faces serious challenges. First, [Fischer & Høyem \(2022\)](#) argue that adjuncts attaching low enough for object control should be process-modifying (e.g. manner adverbials), while those that attach at the vP level and allow subject control should be event-modifying. This account predicts that participants in [Janke & Bailey \(2017\)](#) who gave subject-control responses for temporal adjuncts viewed the adjunct as event-modifying, while those who gave object-control responses viewed it as process-modifying (i.e. the manner in which Harry tapped Luna). This seems unlikely. Furthermore, [Landau \(2021\)](#) points out that V-to-v raising is standardly assumed to occur in all VPs, not just those with low-adjoining adjuncts. Since subject control would therefore also involve V-to-v movement, Janke & Bailey's claim that it is preferred due to its simpler structure is unmotivated.

[Landau \(2021\)](#) provides an alternative account of object control in temporal adjuncts based on his (2015) Two-tiered Theory of Control. He argues only subject control in

temporal adjuncts can involve a syntactic (OC) dependency between PRO and its controller, resulting from the structure in (7a). However, these adjuncts can also exhibit NOC, with either of the structures in (7b–c). In these types of structures, PRO is not syntactically controlled, but is resolved through the discourse, referring either to a logophoric center or an established topic. This theory explains object control as in (4) and (5) with the same mechanism, and predicts that it will be subject to the same constraints as other instances of NOC.

(7) Adjunct control in the Two-tiered Theory of Control (Landau 2021)

(a) *OC adjuncts (predicative control)*:

[<sub>PP</sub> after [<sub>FinP</sub> PRO<sub>i</sub> Fin [<sub>TP</sub> PRO<sub>i</sub> . . . ]]]

(b) *NOC adjuncts (logophoric control)*:

[<sub>PP</sub> after [<sub>CP</sub> *pro* C<sub>+log</sub> [<sub>FinP</sub> PRO<sub>i</sub> Fin [<sub>TP</sub> PRO<sub>i</sub> . . . ]]]]

(c) *NOC adjuncts (topic-bound control)*:

[<sub>PP</sub> after [<sub>CP</sub> *pro* C<sub>+top</sub> [<sub>FinP</sub> PRO<sub>i</sub> Fin [<sub>TP</sub> PRO<sub>i</sub> . . . ]]]]

The structure in (7b) results in logophoric control. Non-subject control of temporal adjuncts has been shown to be more acceptable if PRO is used to refer to some perspective holder in the discourse (Williams 1992; Landau 2015, 2017). For example, in (2), PRO is best understood as referring to the speaker, whose opinion is expressed in the matrix clause. The same is true for (3) on the NOC reading. When PRO cannot refer to the speaker in (8a) due to the third-person reflexive in the adjunct, NOC of PRO becomes degraded unless a third-person perspective holder is directly introduced, as in (8b) (Green 2019b).

- (8) a. ??That oasis was a vision [after PRO dragging himself through the desert all day]. (Green 2019b: 142)
- b. That oasis was a vision to Sam<sub>i</sub> [after PRO<sub>i</sub> dragging himself through the desert all day].

**Table 1.** *Controller Worthiness Scale (Landau 2021).*

|                               | Local subject<br>[+human] | Local subject<br>[-human] | Non-local DP<br>[+human] | Non-local DP<br>[-human] |
|-------------------------------|---------------------------|---------------------------|--------------------------|--------------------------|
| OC-worthy                     | +                         | +                         | -                        | -                        |
| NOC <sub>[+log]</sub> -worthy | +                         | -                         | +                        | -                        |
| NOC <sub>[+top]</sub> -worthy | +                         | +(weak)                   | +(weak)                  | -                        |

In order to account for the preference for subject control and the observed effects of logophoricity and topicality on non-subject control, Landau (2021) argues that the resolution of PRO in temporal and similar adjuncts is restricted by processing constraints in what he calls the Controller Worthiness Scale, given in Table 1. According to this scale, subject control is preferred because it can arise through any of the three strategies to resolve PRO. As the matrix subject c-commanding PRO, it can bind PRO in an OC dependency. If it is a human subject, it can also often serve as a logophoric perspective holder and NOC antecedent for PRO. And as the subject, it is a default discourse topic, also allowing for NOC antecedence. By contrast, because the matrix object does not c-command PRO, object control is predicted to be available only if the object's referent is established as a logophoric perspective holder or discourse topic.

Landau argues that NOC should not be possible with inanimate antecedents, since inanimate objects cannot hold perspective and can only be weakly established as discourse topics (see also Landau 2015, 2017; Green 2019a). Relatedly, Parker, Lago & Phillips (2015) provide evidence that even subject control of temporal adjuncts prefers animate antecedents. However, Donaldson (2021a,b) argues that inanimate NOC controllers are possible based on examples such as (9). Although this involves a different type of adjunct, PRO is still understood to refer to an inanimate, extra-sentential referent, which can only be NOC. Therefore, if NOC can involve inanimate antecedents, and if Landau is correct that object control in temporal adjuncts is an instance of NOC, then PRO in temporal adjuncts should allow inanimate antecedents in matrix object position, even if animate antecedents are preferred.

- (9) [PRO Being made of stainless steel], rust won't be an issue.

However, if object control involves NOC, it should be sensitive to similar interpretive constraints as extra-sentential NOC in temporal adjuncts, as in (2) and (3). NOC accounts such as that of Landau (2021) are therefore challenged by an additional finding of Janke (2018): adults and children in that experiment did not allow extra-sentential controllers in temporal adjuncts, even when they were made highly topical, in contrast to the effect of topicality seen for object control. Janke argues that this evidence favors a structural ambiguity account of object control (rather than NOC-based accounts).

## 1.2. *The current study*

The goals of the current study are as follows: (i) to confirm the results of Janke & Bailey (2017) that object control in temporal adjuncts is not only likely but also acceptable given the proper context, and (ii) to explore what types of context facilitate object control—specifically, what roles animacy and topicality of the antecedent play. In three experiments, we presented participants with adjunct control structures in contexts varying in the strength of their referential bias. In Experiment 1, PRO was biased to corefer with either the matrix subject or object through topicality and perspective, as well as implicit causality (Garvey & Caramazza 1974, Koornneef & Van Berkum 2006, Järvikivi, van Gompel & Hyönä 2017). In Experiment 2, implicit causality alone was used to bias towards subject or object control. Experiment 3 contained no contextual bias toward one controller or the other. In all three experiments, we manipulated the animacy of potential controllers. Participants were asked to choose a likely referent for PRO using a Likert scale and were then asked how natural the sentences were given a specific referent. As a comparison to PRO, we also presented participants with adjuncts containing pronominal subjects whose gender and animacy matched either the matrix subject or object.

If the results of Janke & Bailey (2017) are generalizable beyond the forced-choice task used in their study, we predict that in at least some trials, participants will select the object's referent as PRO's antecedent in the reference choice task, and rate object control interpretations as natural in the acceptability judgment task. Furthermore, we predict



that object control will be more available and acceptable when the object's referent is established as a strong topic (Experiment 1) than when it is not (Experiments 2 and 3). If the Controller Worthiness Scale of Landau (2021) is correct, then in addition to the effect of topicality, we predict an effect of animacy on the resolution of PRO, since only animate referents can be potential logophoric antecedents. An animacy effect is especially likely in Experiment 1, where the animate referent's perspective is explicitly mentioned in the preceding context. If topicality and logophoricity are not required for object control to be available, then we also expect relatively high ratings for object control in Experiment 2, where only implicit causality biases the interpretation.

The experiment protocols for all experiments were approved by the Institutional Review Board at [TO BE ADDED POST-REVIEW]. In each experiment, monolingual English-speaking participants living in the United States were recruited through Prolific (<https://www.prolific.co/>). All participants gave informed consent and were compensated for their time.

## 2. Experiment 1

The purposes of Experiment 1 were (i) to confirm the findings of Janke & Bailey (2017) that object control of temporal adjuncts is available with strong contextual support and (ii) to examine whether animacy of the controller matters, as Landau (2021) suggested. Because Janke & Bailey (2017) used only a forced-choice referent choice task, even when participants selected the matrix object as the controller, they may have done so reluctantly. The matrix object may have seemed the most likely potential referent under discussion even if object control would still be judged ungrammatical. Because of this, we used both a reference choice and acceptability rating task. In addition, all potential controllers in Janke & Bailey's experiments were human and therefore animate. We tested the effects of animacy by including both animate and inanimate potential controllers.

Contextual bias toward a single controller was manipulated in two ways. First, similar to Janke & Bailey (2017), we used background context to establish the target character as

the discourse topic. Second, the matrix clause of the target sentence contained an implicit causality verb (Garvey & Caramazza 1974, Koornneef & Van Berkum 2006, Järvikivi, van Gompel & Hyönä 2017). This verb was intended to strengthen the prediction that the target character would be referred to again in the temporal adjunct clause.

We compared the resolution of PRO to that of overt pronouns that matched in animacy and gender with either the matrix subject or object of the target sentence. Gender was based on stereotypically gendered names of characters as well as pronoun use in the prior context. This comparison served two purposes. First, it helped validate that the context and implicit causality manipulations did indeed bias toward a single referent. If a given pronoun conflicted with the expected referent, we predicted acceptability ratings to be reduced. Second, the pronouns were used to ensure that within the context of the sentences, both characters could still be referred to as the subject of the adjunct clause, despite the contextual bias.

## **2.1. Methods**

### **2.1.1. Participants**

Data was collected from 36 participants (17 female, 9 male; mean age = 39.1, *SD* = 12.5).

### **2.1.2. Stimuli**

Participants viewed stimuli such as those in (10) and (11). Each item consisted of an opening context that established a single character or item as the topic of discourse, as well as the gender/animacy of that topic. In items such as (10) with an animate topic character, the character's perspective was also established. The context was followed by a target sentence containing an implicit causality predicate and a temporal adjunct clause. The implicit causality predicate biased the reader toward interpreting the subject of the temporal adjunct as the topic established in the prior context. Stimuli had a 2×3×2 design: we manipulated implicit causality bias (bias for coreference with the subject or object of

the target sentence), reference form (PRO, subject-oriented pronoun, or object-oriented pronoun), and animacy of the topic-and-bias-consistent referent (animate or inanimate). The term *object* is used loosely here to refer to any non-subject (direct object, indirect object, or object of a preposition). Critical items were followed by three questions: one asking who performed the action in the embedded clause, and two acceptability rating questions based on a specific referent for the subject of the adjunct.

(10) **Context:** Sergio is a big fan of Alyssa and has watched her soccer games since he was a kid. Although they've never met, Sergio recently got better tickets so he could get her attention for the first time.

- (a) **Subject bias:** Sergio got Alyssa's attention after {sitting/he sat/she sat} in a front-row seat.
- (b) **Object bias:** Alyssa noticed Sergio after {sitting/she sat/he sat} in a front-row seat.

### Questions

- (a) Who sat in a front-row seat? (5-point scale from "definitely Sergio" to "definitely Alyssa")
- (b) If it was {Sergio/Alyssa} who sat in a front row seat, how natural would the above sentence sound, given the context? (5-point scale from "completely unnatural" to "completely natural")

(11) **Context:** The orange crayon is the best, because it can make tigers. But it got a little warm in the sun.

- (a) **Subject bias:** The crayon disappointed Edward after {melting/it melted/he melted} in the heat.
- (b) **Object bias:** Edward was disappointed with the crayon after {melting/he melted/it melted} in the heat.

### Questions

- (a) Who or what melted in the heat? (5-point scale from “definitely the crayon” to “definitely Edward”)
- (b) If it was {Edward/the crayon} that melted in the heat, how natural would the above sentence sound, given the context? (5-point scale from “completely unnatural” to “completely natural”)

Stimuli included a total of 48 sets of items (24 each in animate and inanimate conditions), distributed across six lists in a Latin-square design, such that each participant saw four sentences in each condition. Each list also contained 50 fillers, for a total of 98 items per list. Fillers were either well-formed or contained morphosyntactic errors, semantic anomalies, or ambiguous pronouns, and were followed by one to three comprehension, reference resolution, or acceptability rating questions.

### 2.1.3. Procedure

Participants completed the experiment on the PCIBex platform ([Zehr & Schwarz 2023](#)). For each item, the context was presented first. After reading the context, participants pressed a button to see the target sentence, which appeared in bold below the context. The participant then pressed a button to see the first question, which appeared below the target sentence, with possible answers given as a five-point Likert scale. After answering one question, the next appeared. Although only one question appeared on the screen at a time, the context and target sentence remained on the screen for each question. After answering the final question, the context for the next item appeared. Items were presented in a randomized order for each participant.

### 2.1.4. Data processing and analysis

Responses to the reference choice and acceptability questions were analyzed separately. For reference choice, responses were coded numerically and z-scored for each participant, such that positive and negative values meant that coreference of the subject of the adjunct with the

matrix subject or object was more likely. A linear mixed-effects model was run to measure the effect of reference form (PRO, subject-oriented pronoun, or object oriented pronoun), bias (toward coreference with the matrix subject or object), and animacy of the biased referent, on reference ratings. These fixed effects were all sum coded (for form, contrast levels included PRO vs. subject-oriented pronoun and PRO vs. object-oriented pronoun). The maximal fixed and random effect structures were determined using the *buildmer* package (Voeten 2023) in R (R Core Team 2024, Posit Team 2025), which orders model terms based on their contribution to the model before performing backward elimination using likelihood-ratio tests. Degrees of freedom were calculated using the Satterthwaite method (Kuznetsova, Brockhoff & Christensen 2017). The final model included all main effects and two-way interactions, as well as random intercepts for items and random slopes by item for bias. The three-way interaction was not included because it did not improve model fit.

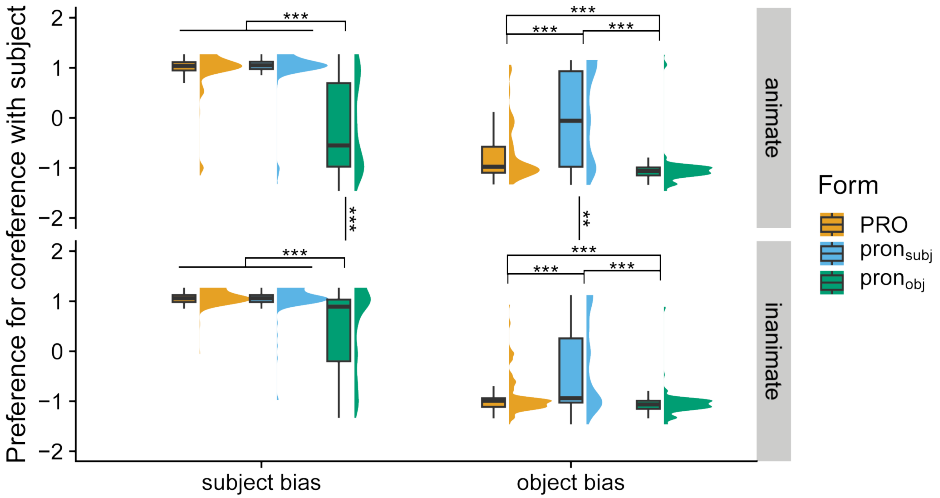
Acceptability judgment responses were also coded numerically and z-scored for each participant. Fixed effects for this model included the same fixed effects as for the reference choice question, as well as a fixed effect for the referent given in the question (sum coded). The maximal feasible model that converged included all main effects and two- and three-way interactions (but no four-way interaction), as well as random intercepts for items and random slopes by item for referent.

For both models, p-values were adjusted to control for the false discovery rate (Benjamini & Hochberg 1995). Tukey-adjusted pairwise post-hoc comparisons were computed to follow up on significant interactions using the *emmeans* package (Lenth 2024). The full data analysis scripts and results are given in the linked OSF repository.

## 2.2. Results

### 2.2.1. Reference choice

The average preference for coreference with the subject in response to the reference resolution question for each condition is illustrated in Figure 1, and the results of the linear



**Figure 1.** Experiment 1: Mean likelihood for critical referential expression to co-refer with the subject of the sentence (z-scored)

**Table 2.** Experiment 1: Fixed effects for reference choice responses..

| Fixed effect  | $\beta$ | SE   | df   | t      | p     |     |
|---------------|---------|------|------|--------|-------|-----|
| (Intercept)   | 0.00    | 0.02 | 46   | 0.00   | 1.000 |     |
| Bias          | -0.69   | 0.03 | 46   | -23.18 | <.001 | *** |
| Form1         | 0.38    | 0.02 | 1626 | 20.23  | <.001 | *** |
| Form2         | -0.46   | 0.02 | 1626 | -24.45 | <.001 | *** |
| Animacy       | -0.02   | 0.02 | 46   | -1.25  | .240  |     |
| Bias:Form1    | 0.09    | 0.02 | 1626 | 4.67   | <.001 | *** |
| Bias:Form2    | 0.12    | 0.02 | 1626 | 6.24   | <.001 | *** |
| Bias:Animacy  | 0.11    | 0.03 | 46   | 3.75   | .001  | *** |
| Form1:Animacy | 0.09    | 0.02 | 1626 | 4.75   | <.001 | *** |
| Form2:Animacy | -0.11   | 0.02 | 1626 | -5.86  | <.001 | *** |

Note: Form1 = pron<sub>subj</sub> vs. PRO; Form2 = pron<sub>obj</sub> vs. PRO. *p*-values adjusted for FDR. \**p* < .05, \*\**p* < .01, \*\*\**p* < .001

mixed-effects model are given in Table 2. We found significant two-way interactions between bias and form, bias and animacy, and form and animacy.

Follow-up comparisons indicated that contextual bias affected all three referent forms. More interpretations included coreference with the matrix subject when context biased toward that interpretation, but the effect was largest for PRO (PRO:  $\beta = -1.79$ , *t*.ratio =

$-25.42, p < .001$ ;  $\text{pron}_{\text{subj}}: \beta = -1.20, t.\text{ratio} = -17.07, p < .001$ ;  $\text{pron}_{\text{obj}}: \beta = -1.14, t.\text{ratio} = -16.23, p < .001$ ).

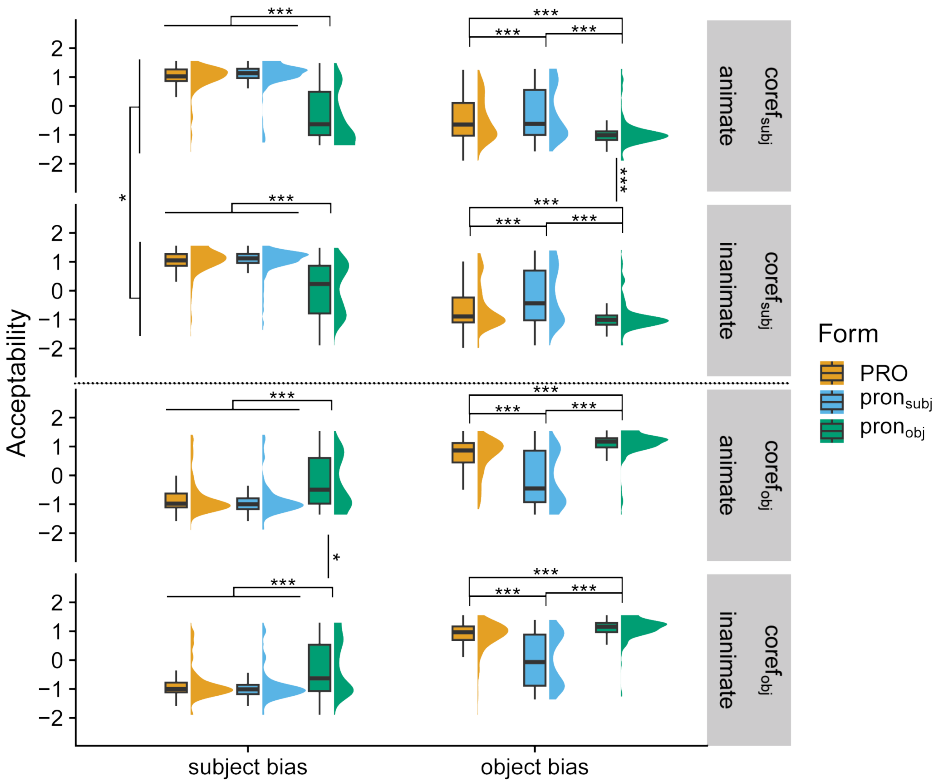
PRO behaved like a subject-oriented pronoun when bias favored coreference with the subject, resulting in similar judgments for the two forms ( $\beta = 0.01, t.\text{ratio} = 0.18, p = .983$ ). In object-biased items, PRO behaved more like an object-oriented pronoun, resulting in greater coreference with the object than was seen for subject-oriented pronouns ( $\beta = 0.60, t.\text{ratio} = 12.90, p < .001$ ). However, coreference with the object was not favored as strongly for PRO as with object-oriented pronouns ( $\beta = -0.22, t.\text{ratio} = -4.71, p < .001$ ).

The interaction between animacy and bias indicated that bias had a stronger effect for inanimate items ( $\beta = -1.60, t.\text{ratio} = -19.05, p < .001$ ) than for animate items ( $\beta = -1.16, t.\text{ratio} = -13.74, p < .001$ ). However, the interaction between animacy and form indicated that although responses to pronouns were affected by animacy ( $\text{pron}_{\text{subj}}: \beta = 0.14, t.\text{ratio} = 2.91, p = .004$ ;  $\text{pron}_{\text{obj}}: \beta = -0.26, t.\text{ratio} = -5.35, p < .001$ ), responses to PRO were not ( $\beta < 0.01, t.\text{ratio} = 0.08, p = .939$ ).

### 2.2.2. Acceptability judgments

Mean normalized acceptability ratings are illustrated in Figure 2. The results for the linear mixed-effects model examining acceptability ratings are given in Table 3. The following three-way interactions were significant:  $\text{form} \times \text{animacy} \times \text{bias}$ ,  $\text{animacy} \times \text{referent} \times \text{bias}$ ,  $\text{form} \times \text{animacy} \times \text{referent}$ , and  $\text{form} \times \text{referent} \times \text{bias}$ .

Follow-up comparisons to these interactions indicated that when context biased toward coreference with the matrix subject, PRO received similar naturalness ratings as a pronoun matching the subject in gender. Coreference with the subject (in line with bias) was rated as similarly natural ( $\beta = 0.06, z.\text{ratio} = 0.97, p = .593$ ), and coreference with the matrix object as similarly unnatural ( $\beta = -0.07, z.\text{ratio} = -1.17, p = .470$ ). Subject-oriented pronouns and PRO differed significantly from object-oriented pronouns, both for acceptability of coreference with the subject ( $\text{pron}_{\text{obj}}$  vs.  $\text{pron}_{\text{subj}}: \beta = 1.13, z.\text{ratio} = 19.97$ ,



**Figure 2.** Experiment 1: Mean acceptability ratings (z-scored)

$p < .001$ ; vs. PRO:  $\beta = -1.08$ ,  $z.ratio = -19.00$ ,  $p < .001$ ), and of coreference with the object (pron<sub>obj</sub> vs. pron<sub>subj</sub>:  $\beta = -0.55$ ,  $z.ratio = -9.76$ ,  $p < .001$ ; vs. PRO:  $\beta = 0.49$ ,  $z.ratio = 8.59$ ,  $p < .001$ ).

When context biased toward coreference with the matrix object, on the other hand, PRO was rated more similarly to an object-oriented pronoun. With coreference to the object (in line with bias), a pronoun matching in gender was rated higher than PRO ( $\beta = 0.32$ ,  $z.ratio = 5.57$ ,  $p < .001$ ), but PRO was rated much higher than a subject-oriented pronoun ( $\beta = -0.80$ ,  $z.ratio = -14.18$ ,  $p < .001$ ). For coreference with the subject (contrary to bias), ratings were reversed, with subject-oriented pronouns receiving higher ratings than PRO ( $\beta = 0.28$ ,  $z.ratio = 4.93$ ,  $p < .001$ ), which in turn received higher ratings than object-oriented pronouns ( $\beta = -0.36$ ,  $z.ratio = -6.37$ ,  $p < .001$ ).



**Table 3.** Experiment 1: Fixed effects for acceptability rating responses..

| Fixed effect           | $\beta$ | SE   | df   | t      | p     |     |
|------------------------|---------|------|------|--------|-------|-----|
| (Intercept)            | -0.00   | 0.02 | 46   | -0.00  | 1.000 |     |
| Form1                  | -0.04   | 0.02 | 3342 | -2.22  | .048  | *   |
| Form2                  | -0.06   | 0.02 | 3342 | -3.74  | .001  | *** |
| Animacy                | -0.02   | 0.02 | 46   | -1.15  | .331  |     |
| Referent               | -0.02   | 0.01 | 46   | -1.65  | .158  |     |
| Bias                   | 0.01    | 0.01 | 3342 | 1.16   | .331  |     |
| Form1:Referent         | -0.39   | 0.02 | 3342 | -23.68 | <.001 | *** |
| Form2:Referent         | 0.47    | 0.02 | 3342 | 28.95  | <.001 | *** |
| Animacy:Referent       | 0.02    | 0.01 | 46   | 1.08   | .349  |     |
| Referent:Bias          | 0.59    | 0.01 | 3342 | 51.17  | <.001 | *** |
| Form1:Bias             | -0.13   | 0.02 | 3342 | -8.01  | <.001 | *** |
| Form2:Bias             | 0.13    | 0.02 | 3342 | 8.17   | <.001 | *** |
| Form1:Animacy          | 0.00    | 0.02 | 3342 | 0.14   | .929  |     |
| Form2:Animacy          | -0.03   | 0.02 | 3342 | -1.61  | .158  |     |
| Animacy:Bias           | -0.01   | 0.01 | 3342 | -0.44  | .722  |     |
| Form1:Referent:Bias    | -0.09   | 0.02 | 3342 | -5.28  | <.001 | *** |
| Form2:Referent:Bias    | -0.07   | 0.02 | 3342 | -4.13  | <.001 | *** |
| Form1:Animacy:Referent | -0.01   | 0.02 | 3342 | -0.91  | .422  |     |
| Form2:Animacy:Referent | 0.05    | 0.02 | 3342 | 3.17   | .003  | **  |
| Animacy:Referent:Bias  | -0.04   | 0.01 | 3342 | -3.50  | .001  | **  |
| Form1:Animacy:Bias     | -0.03   | 0.02 | 3342 | -2.09  | .062  |     |
| Form2:Animacy:Bias     | 0.04    | 0.02 | 3342 | 2.31   | .042  | *   |

Note: Form1 = pron<sub>subj</sub> vs. PRO; Form2 = pron<sub>obj</sub> vs. PRO. *p*-values adjusted for FDR. \**p* < .05, \*\**p* < .01, \*\*\**p* < .001

Similar findings were revealed when directly comparing ratings based on coreference within each form. When bias favored coreference with the subject, subject coreference received higher ratings than object coreference (pron<sub>subj</sub>:  $\beta = -1.83$ ,  $z.ratio = -30.83$ ,  $p < .001$ ; pron<sub>obj</sub>:  $\beta = -0.15$ ,  $z.ratio = -2.52$ ,  $p = .012$ ; PRO:  $\beta = -1.72$ ,  $z.ratio = 28.79$ ,  $p < .001$ ). When bias favored coreference with the object, object coreference received higher ratings than subject coreference (pron<sub>subj</sub>:  $\beta = 0.18$ ,  $z.ratio = 3.16$ ,  $p = .002$ ; pron<sub>obj</sub>:  $\beta = 1.95$ ,  $z.ratio = 32.74$ ,  $p < .001$ ; PRO:  $\beta = 1.27$ ,  $z.ratio = 21.36$ ,  $p < .001$ ). However, the effect of bias was strongest for PRO.

The interactions with animacy were driven by effects for pronouns, especially those with the object's gender. Object-oriented pronouns received higher ratings for inanimate items than for animate ones when their gender conflicted with contextual bias ( $\beta = -0.17$ ,

$z.ratio = -2.49, p = .013$ ). Similarly, object-oriented pronouns resulted in higher ratings for coreference with the subject in inanimate items than animate ones ( $\beta = -0.24, z.ratio = -3.44, p < .001$ ). The effect of animacy based on bias for subject-oriented pronouns was only marginal ( $\beta = -0.12, z.ratio = -1.84, p = .067$ ), and all animacy effects for PRO were insignificant (all  $p > .5$ ).

The three-way interaction between animacy, referent, and bias indicated that with bias toward the subject, inanimate items were rated higher for coreference with the subject than animate ones, regardless of reference form ( $\beta = -0.15, z.ratio = -2.50, p = .012$ ).

### **2.3. Discussion**

Janke & Bailey (2017) found that object control of temporal adjuncts is more available when the object is strongly primed as a discourse topic, with roughly half of responses indicating object control interpretations. However, it is unclear how acceptable object control was for participants given the forced-choice nature of their task—participants may have selected the object as a more likely antecedent to PRO than the subject, despite such an interpretation being ungrammatical for them.

This experiment confirms and extends Janke & Bailey's findings. The reference choice results indicate that participants found the object to be the most likely antecedent to PRO when context biased toward that interpretation. And even though participants could indicate in the Likert scale that PRO could corefer with both the subject and the object, responses indicate that they overwhelmingly favored object control over subject control. In addition, acceptability ratings indicate that object control with object bias was not only available, but highly acceptable. However, coreference with the matrix object was not as strongly favored as for overt pronouns matching in gender with the object. This suggests that participants still recognized that subject control was possible, and occasionally preferred it. Even so, subject control in object-biased items was generally rated as unnatural, suggesting that the context was able to override the grammatical preference for subject control enough to make it less acceptable.

Unsurprisingly, subject control was strongly preferred when it was supported by contextual bias, as evidenced by both reference choice and acceptability ratings. In fact, when context biased toward coreference with the subject, there was no difference in responses for PRO and overt pronouns matching the subject in gender. This suggests that participants did not consider object control at all. This contrasts with the somewhat weaker preference for object control in object-biased items.

This experiment also examined the effect of animacy of potential antecedents on control preferences. Previous research found evidence that PRO in temporal adjuncts prefers animate antecedents (Kawasaki 1993, Landau 2001, Parker, Lago & Phillips 2015). This potential preference would be consistent with Landau's (2021) prediction that NOC (any instance of non-subject control in temporal adjuncts) should be more available for animate than inanimate potential controllers. However, the only effect of animacy in this experiment was for overt pronouns; participants allowed non-gender-agreeing pronouns more often for inanimate referents than for animate ones. This is likely due to the possibility of referring to inanimate objects in English as "she" or "he", such as when someone personifies their car by saying, "She's a beauty." Crucially, we found no effect of animacy on PRO items, even for subject-control readings. This result presents a potential challenge to Landau's Controller Worthiness Scale, which claims that inanimate non-subject controllers should be virtually impossible in temporal adjuncts (and that animate subject controllers should be preferred over inanimate ones).

Although this experiment demonstrated that object control of temporal adjuncts can be readily available and acceptable, it did so only with a very strongly supporting context. In object-biased sentences, the prior context established the relevant referent as a strong context, and in animate items, we attempted to highlight that referent's perspective in order to facilitate logophoric antecedence. It is therefore unclear how strong the contextual support needs to be to allow object control interpretations. In addition, the strong topic bias may have masked animacy effects that really were present. Experiment 2 examines the availability of object control in temporal adjuncts in contexts with reduced bias.

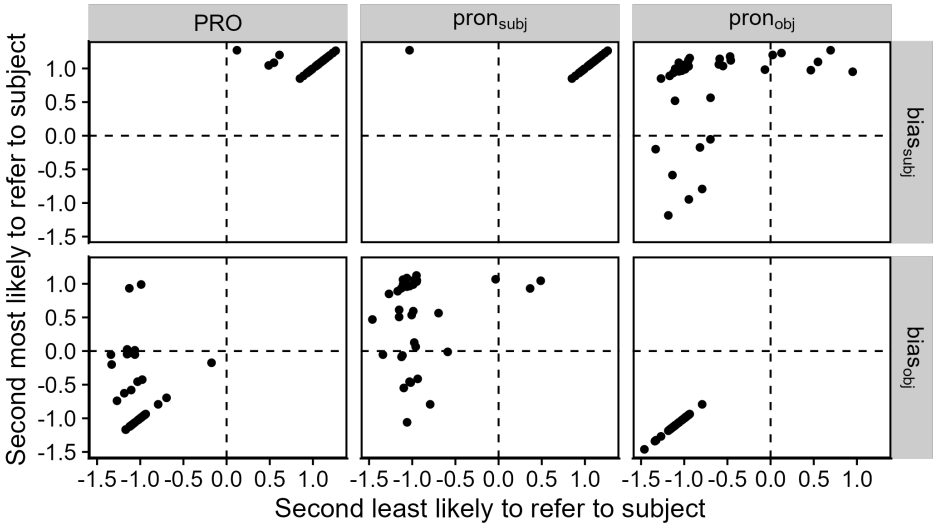
### 2.3.1. Exploratory analyses

In addition to contextual effects on PRO, overt pronouns were also affected by bias, despite the prior context using pronouns to establish the characters' gender. However, the responses to pronouns whose gender mismatched the biased referent showed a broad, somewhat bimodal distribution. This suggests that there were inconsistencies in responses, either across participants or across items. To investigate whether this was due to individual differences across participants, we examined individual responses following [Huang \(2024\)](#), who adapted similar methods from [Kush, Lohndal & Sprouse \(2019\)](#) and [Pañeda & Kush \(2021\)](#). For each condition, the highest and lowest ratings from each participant were removed to exclude any potential outliers. The remaining second-highest and second-lowest ratings were then plotted, with a separate dot representing each participant (for simplicity, we removed the animacy distinction for this exploratory analysis). The plot for reference choice is given in [Figure 3](#). Participants who consistently chose coreference with the subject for a given condition are located in the top right quadrant for that condition. Participants who consistently chose coreference with the object are found in the bottom left quadrant. Participants who sometimes preferred coreference with the subject and other times preferred coreference with the object are found in the top left quadrant.

Pronouns conflicting with bias induced wide variability in responses. Some participants consistently favored the referent matching in gender with the pronoun, some consistently chose the referent matching the context bias, and some were inconsistent in their responses.

Importantly, there was less variability with object control of PRO. Nearly everyone consistently favored object control when context biased towards it, or allowed both subject and object control. The few participants who strongly preferred subject control on at least one item still strongly preferred object control on at least one other.

Similar results are seen in acceptability ratings ([Figure 4](#)). Although ratings varied when reference contrasted with contextual bias, most participants consistently rated object control in object-biased sentences as acceptable, almost to a similar extent as with subject control in subject-biased sentences. Participants who rated object control as unnatural in

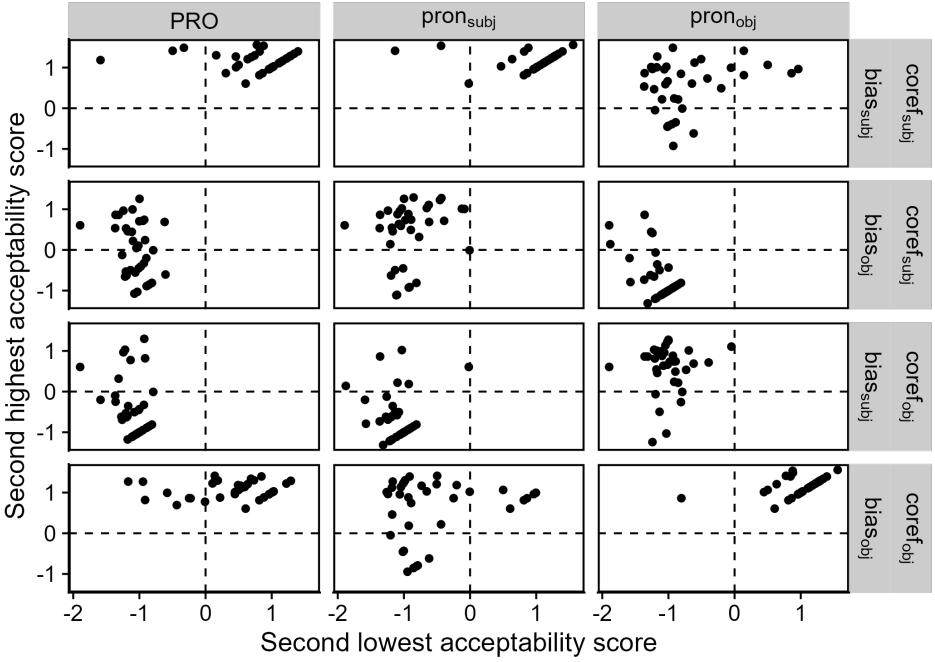


**Figure 3.** Experiment 1: Subject preferences by participant (each dot = one participant)

at least some sentences still found it highly acceptable in others. These results suggest that object control of PRO was broadly preferred and judged as acceptable across speakers when favored by a strongly-supporting context.

### 3. Experiment 2

Experiment 2 reduced the strength of the contextual bias by removing the initial context sentence. This meant that object control would not be supported by strong topicality or logophoricity. Instead, referential bias was solely dependent on the implicit causality in the main clause for each item. It was predicted that object control would still be possible, but perhaps less preferred than in Experiment 1. In addition, if the lack of animacy effect seen in Experiment 1 was due to masking of the effect by the strength of the contextual bias, then reducing that bias might uncover an animacy effect.



**Figure 4.** *Experiment 1: Acceptability by participant (each dot = one participant)*

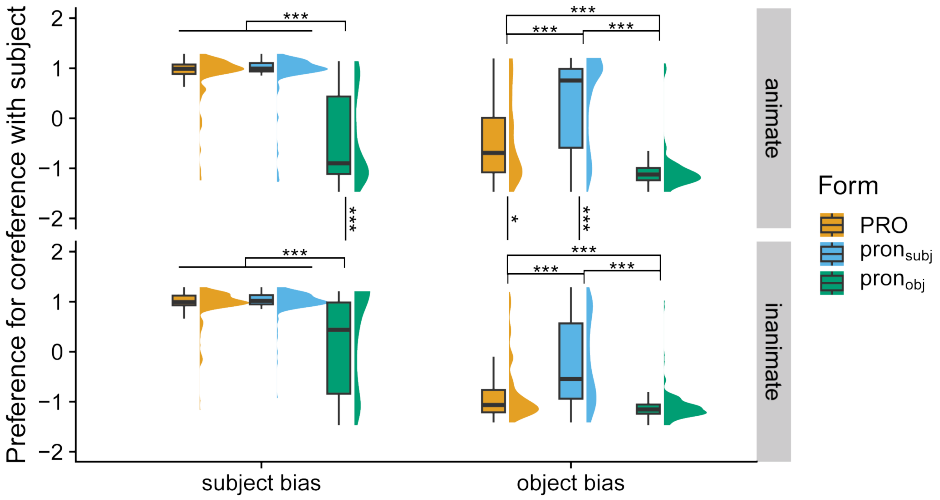
### 3.1. Methods

#### 3.1.1. Participants

Data was collected from 36 participants (23 female, 12 male, 1 preferred not to say; mean age = 39.9,  $SD = 13.2$ ). Data from one participant was corrupted, so the final data set contained 35 participants.

#### 3.1.2. Materials, procedure, and analysis

The stimuli and procedure for Experiment 2 were similar to those used in Experiment 1, except that the initial context sentence was not shown to participants. The data analysis replicated that of Experiment 1. The final model for reference choice again included main effects of bias, form, and animacy, all two-way interactions, as well as random intercepts for items with random slopes by item for bias. For acceptability ratings, the final model included main effects for bias, form, animacy, and referent; all two-way interactions; and



**Figure 5.** Experiment 2: Mean likelihood for critical referential expression to co-refer with the subject of the sentence (z-scored)

three-way interactions for form  $\times$  referent  $\times$  bias and animacy  $\times$  referent  $\times$  bias. Since the model did not converge with random effects, a basic linear model was used.

### 3.2. Results

#### 3.2.1. Reference choice

The average preference for coreference with the subject in response to the reference resolution question for each condition is illustrated in Figure 5, and the results of the linear mixed-effects model are given in Table 4. Similar to Experiment 1, all of the two-way interactions were significant.

As in Experiment 1, follow-up comparisons indicated that all three referent forms were affected by contextual bias, with greater coreference with the subject when context biased toward that interpretation, but PRO items showed the largest effect of bias ( $\text{pron}_{\text{subj}}$ :  $\beta = -0.91$ ,  $t.\text{ratio} = -13.26$ ,  $p < .001$ ;  $\text{pron}_{\text{obj}}$ :  $\beta = -0.90$ ,  $t.\text{ratio} = -13.16$ ,  $p < .001$ ; PRO:  $\beta = -1.51$ ,  $t.\text{ratio} = -22.14$ ,  $p < .001$ ).

**Table 4.** *Experiment 2: Fixed effects for reference choice responses.*

| Fixed effect  | $\beta$ | <i>SE</i> | <i>df</i> | <i>t</i> | <i>p</i> |     |
|---------------|---------|-----------|-----------|----------|----------|-----|
| (Intercept)   | 0.00    | 0.02      | 46        | 0.00     | .999     |     |
| Bias          | -0.55   | 0.03      | 46        | -20.72   | <.001    | *** |
| Form1         | 0.49    | 0.02      | 1580      | 22.92    | <.001    | *** |
| Form2         | -0.61   | 0.02      | 1580      | -28.62   | <.001    | *** |
| Animacy       | 0.01    | 0.02      | 45.99     | 0.72     | .528     |     |
| Bias:Form1    | 0.10    | 0.02      | 1580      | 4.69     | <.001    | *** |
| Bias:Form2    | 0.10    | 0.02      | 1580      | 4.85     | <.001    | *** |
| Bias:Animacy  | 0.14    | 0.03      | 46        | 5.39     | <.001    | *** |
| Form1:Animacy | 0.09    | 0.02      | 1580      | 4.16     | <.001    | *** |
| Form2:Animacy | -0.15   | 0.02      | 1580      | -6.96    | <.001    | *** |

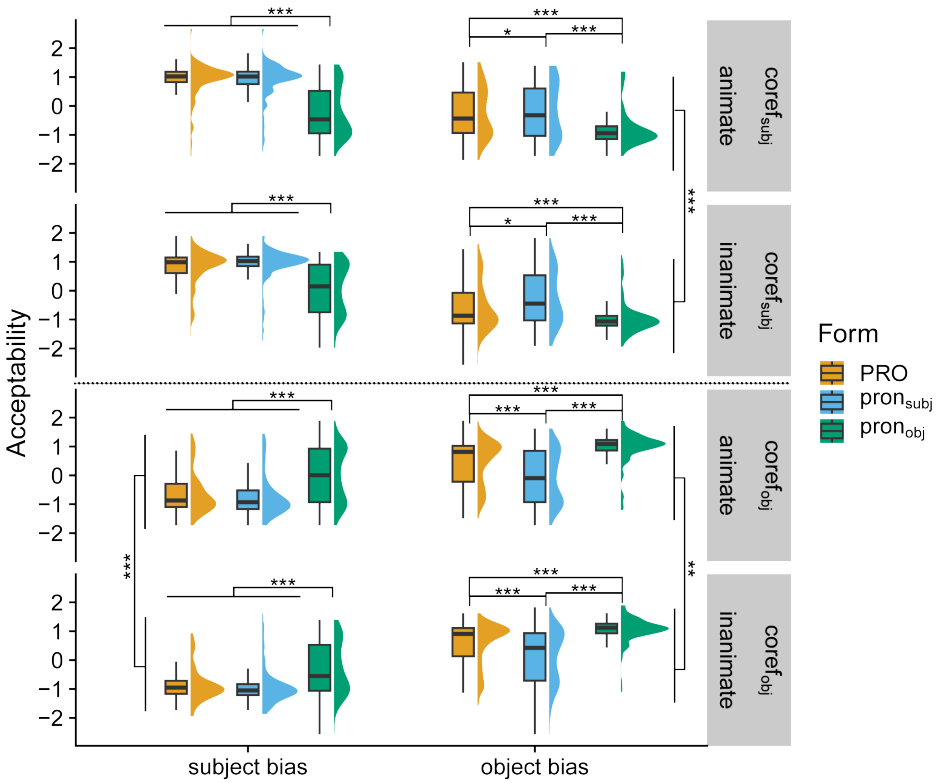
*Note:* Form1 = pron<sub>subj</sub> vs. PRO; Form2 = pron<sub>obj</sub> vs. PRO. *p*-values adjusted for FDR. \**p* < .05, \*\**p* < .01, \*\*\**p* < .001

PRO once again behaved like a subject-oriented pronoun when bias favored coreference with the subject, resulting in no statistical difference in judgments between the two forms ( $\beta = 0.06$ ,  $t_{ratio} = 1.22$ ,  $p = .442$ ). And as in Experiment 1, in object-biased items, PRO behaved more like an object-oriented pronoun, resulting in greater coreference with the object than was seen for subject-oriented pronouns ( $\beta = 0.67$ ,  $t_{ratio} = 12.83$ ,  $p < .001$ ). However, coreference with the object was again not favored as strongly as with object-oriented pronouns ( $\beta = -0.43$ ,  $t_{ratio} = -8.14$ ,  $p < .001$ ).

Overall, subject-oriented pronouns and PRO resulted in higher coreference with the subject when that subject was animate than when it was inanimate (pron<sub>subj</sub>:  $\beta = 0.21$ ,  $t_{ratio} = 3.54$ ,  $p < .001$ ; PRO:  $\beta = 0.15$ ,  $t_{ratio} = 2.54$ ,  $p = .012$ ). Similarly, object-oriented pronouns were more likely to corefer with animate objects ( $\beta = -0.27$ ,  $t_{ratio} = -4.61$ ,  $p < .001$ ). Because there was no three-way interaction with bias, however, it is unclear whether this was true across the board or whether it was affected by bias. However, visual inspection of the results suggests that the effect was driven by items with object bias.

Averaging over reference forms, bias toward the subject or object had a stronger effect—leading to more and less frequent coreference with the subject, respectively—for inanimate items than for animate items (animate:  $\beta = -0.82$ ,  $t_{ratio} = -10.84$ ,  $p < .001$ ; inanimate:  $\beta = -1.40$ ,  $t_{ratio} = -18.46$ ,  $p < .001$ ).





**Figure 6.** Experiment 2: Mean acceptability ratings (z-scored)

### 3.2.2. Acceptability judgments

Mean normalized acceptability ratings are illustrated in Figure 6. The results for the linear model examining acceptability ratings are given in Table 5. There were significant three-way interactions for form  $\times$  referent  $\times$  bias and animacy  $\times$  referent  $\times$  bias.

As in Experiment 1, follow-up comparisons indicated that when the context biased toward coreference with the matrix subject, PRO received similar naturalness ratings as subject-oriented pronouns. Again, coreference with the subject, in line with bias, was rated as similarly natural for the two forms ( $\beta = 0.05$ ,  $t_{ratio} = 0.80$ ,  $p = .705$ ), and coreference with the matrix object as similarly unnatural ( $\beta = -0.09$ ,  $t_{ratio} = -1.37$ ,  $p = .355$ ). Subject-oriented pronouns and PRO again differed significantly from object-oriented pronouns, both for acceptability of coreference with the subject (pron<sub>obj</sub> vs.

**Table 5.** *Experiment 2: Fixed effects for acceptability rating responses.*

| Fixed effect          | $\beta$ | SE   | $t$    | $p$   |     |
|-----------------------|---------|------|--------|-------|-----|
| (Intercept)           | -0.00   | 0.01 | -0.00  | 1.000 |     |
| Form1                 | -0.03   | 0.02 | -1.62  | .153  |     |
| Form2                 | -0.03   | 0.02 | -1.40  | .208  |     |
| Animacy               | 0.03    | 0.01 | 2.37   | .036  | *   |
| Referent              | -0.02   | 0.01 | -1.89  | .093  |     |
| Bias                  | 0.01    | 0.01 | 0.69   | .560  |     |
| Form1:Referent        | -0.33   | 0.02 | -18.06 | <.001 | *** |
| Form2:Referent        | 0.47    | 0.02 | 25.30  | <.001 | *** |
| Referent:Bias         | 0.53    | 0.01 | 40.31  | <.001 | *** |
| Form1:Bias            | -0.08   | 0.02 | -4.36  | <.001 | *** |
| Form2:Bias            | 0.09    | 0.02 | 5.07   | <.001 | *** |
| Animacy:Bias          | -0.02   | 0.01 | -1.38  | .208  |     |
| Animacy:Referent      | 0.00    | 0.01 | 0.07   | 1.000 |     |
| Form1:Referent:Bias   | -0.04   | 0.02 | -2.41  | .036  | *   |
| Form2:Referent:Bias   | -0.04   | 0.02 | -2.23  | .046  | *   |
| Animacy:Referent:Bias | -0.08   | 0.01 | -6.28  | <.001 | *** |

*Note:* Form1 = pron<sub>subj</sub> vs. PRO; Form2 = pron<sub>obj</sub> vs. PRO.  $p$ -values adjusted for FDR. \* $p < .05$ , \*\* $p < .01$ , \*\*\* $p < .001$

pron<sub>subj</sub>:  $\beta = 0.97$ ,  $t_{ratio} = 15.12$ ,  $p < .001$ ; vs. PRO:  $\beta = -0.92$ ,  $t_{ratio} = -14.32$ ,  $p < .001$ ), and of coreference with the object (pron<sub>obj</sub> vs. vs. pron<sub>subj</sub>:  $\beta = -0.63$ ,  $t_{ratio} = -9.81$ ,  $p < .001$ ; vs. PRO:  $\beta = 0.54$ ,  $t_{ratio} = 8.43$ ,  $p < .001$ ).

Also as in Experiment 1, when context biased toward coreference with the matrix object, PRO was rated more similarly to an object-oriented pronoun. With coreference to the object, in line with bias, a pronoun matching in gender was rated higher than PRO ( $\beta = 0.50$ ,  $t_{ratio} = 7.80$ ,  $p < .001$ ), but PRO was rated higher than a subject-oriented pronoun ( $\beta = -0.49$ ,  $t_{ratio} = -7.55$ ,  $p < .001$ ). For coreference with the subject, ratings were again reversed, with subject-oriented pronouns receiving higher ratings than PRO ( $\beta = 0.18$ ,  $t_{ratio} = 2.77$ ,  $p = .016$ ), which received higher ratings than object-oriented pronouns ( $\beta = -0.45$ ,  $t_{ratio} = -7.01$ ,  $p < .001$ ).

When directly comparing ratings based on coreference within each form, findings were again similar to Experiment 1 for subject-oriented pronouns and PRO, but not for object-oriented pronouns. When bias favored coreference with the subject, subject coreference received higher ratings than object coreference except for with object-oriented pronouns

(PRO:  $\beta = -1.55$ ,  $t.ratio = -24.09$ ,  $p < .001$ ; pron<sub>subj</sub>:  $\beta = -1.69$ ,  $t.ratio = -26.26$ ,  $p < .001$ ; pron<sub>obj</sub>:  $\beta = -0.09$ ,  $t.ratio = -1.34$ ,  $p = .182$ ). When bias favored coreference with the object, object coreference received higher ratings than subject coreference for all three forms (PRO:  $\beta = 0.91$ ,  $t.ratio = 14.19$ ,  $p < .001$ ; pron<sub>subj</sub>:  $\beta = 0.25$ ,  $t.ratio = 3.86$ ,  $p < .001$ ; pron<sub>obj</sub>:  $\beta = 1.86$ ,  $t.ratio = 29.00$ ,  $p < .001$ ).

The interaction with animacy indicated that when bias favored coreference with the subject, object coreference was more acceptable for animate than for inanimate referents ( $\beta = 0.26$ ,  $t.ratio = 5.05$ ,  $p < .001$ ). Subject coreference was more acceptable for animate items when bias favored object coreference ( $\beta = 0.19$ ,  $t.ratio = 3.61$ ,  $p < .001$ ). But when bias favored object coreference, object coreference was more acceptable for inanimate objects ( $\beta = -0.14$ ,  $t.ratio = -2.61$ ,  $p = .009$ ). There was no evidence for an effect of animacy when both bias and coreference favored the subject ( $\beta = -0.07$ ,  $t.ratio = -1.30$ ,  $p = .193$ ).

### 3.3. Discussion

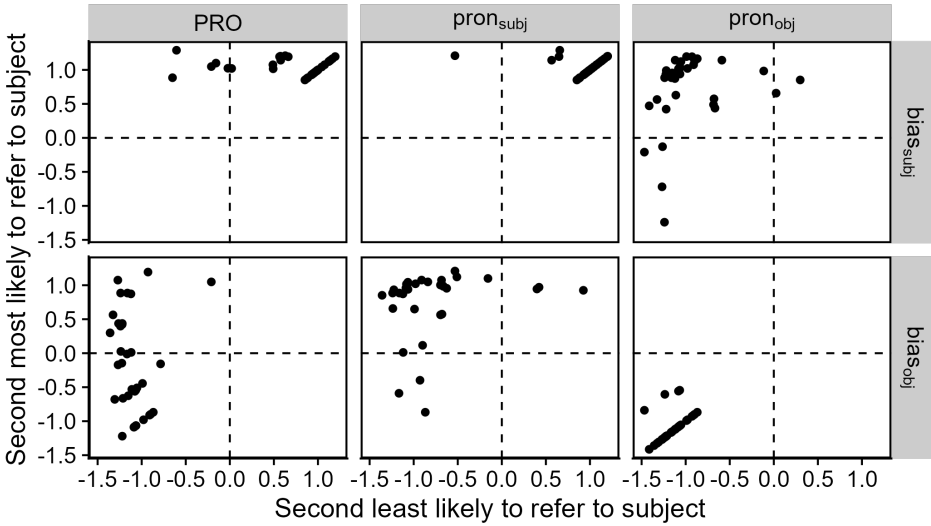
Similar to Experiment 1's findings, PRO items were strongly affected by contextual bias for both reference choice and acceptability. PRO received similar interpretations and acceptability judgments as a subject-oriented pronoun with subject bias only. With object bias, PRO behaved more similarly to an object-oriented pronoun. Once again, though, the preference for object coreference was not quite as strong for PRO as for an overt pronoun. Therefore, the results of this experiment demonstrate that even without a prior sentence establishing a topic of discourse, object control is still available in temporal adjuncts with object-biased implicit causality.

In contrast to Experiment 1, animacy did have a significant effect on PRO in Experiment 2. On reference choice questions, PRO had a larger proportion of subject-control interpretations for animate controllers than for inanimate controllers. This is consistent with previous results that found that adjunct control is more acceptable with animate (subject)

antecedents (Kawasaki 1993, Landau 2001, Parker, Lago & Phillips 2015), and is consistent with Landau's 2021 Controller Worthiness Scale. However, the Controller Worthiness Scale also claims that [+human] DPs should allow NOC, and therefore object control, but that inanimate DPs should not (or at least be less likely to). This was not the case in this experiment, as object control was more strongly preferred with inanimate antecedents than with animate ones. This may be due in part to the fact that in object-biased items, the implicit causality verbs often expressed the subject character's perspective. This may have discouraged logophoric object control and therefore limited the effect of animacy. However, animacy did not have an effect at all on object control in Experiment 1, when prior context established the object's perspective in animate items. This may indicate that logophoric object control was more available for animate items Experiment 1, reducing the bias for subject control.

The effect of animacy on acceptability ratings contrasted somewhat with its effect on reference choice. While reference choice indicated a stronger subject-bias for animate controllers, animate items did not consistently receive higher acceptability ratings for subject coreference than for object coreference. Instead, animacy led to an advantage any time reference resolution conflicted with bias; animate *object* antecedents received higher ratings than inanimate ones when bias was toward the subject. In addition, when bias was toward the object, *inanimate* items received higher ratings overall than animate ones. Because there was no interaction with form, however, this appears to have been true across all forms, not just PRO, suggesting a similar effect of animacy on PRO and overt pronouns.

Landau's (2021) Controller Worthiness Scale states that NOC—including any instance of non-subject control in temporal adjuncts—should be most likely with logophoric antecedents, somewhat likely with strongly topical antecedents, and highly unlikely with inanimate antecedents. Importantly, in this experiment, object control was available even though the object of the sentence was neither a topic nor a logophoric center, but was simply the most likely referent due to implicit causality. These results suggest that logophoricity or topicality are not necessary conditions for object control. Instead, resolution of object



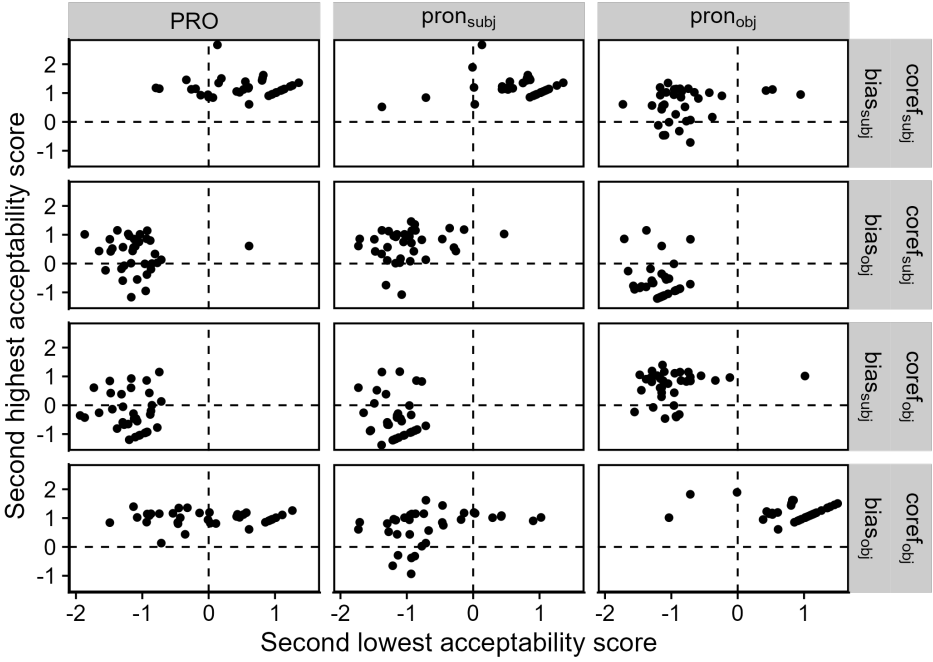
**Figure 7.** Experiment 2: Subject preferences by participant (each dot = one participant)

control in temporal adjuncts, and perhaps of NOC more generally, may be largely reducible to resolution of null pronouns. We will return to this possibility in the general discussion.

### 3.3.1. Exploratory analyses

As in Experiment 1, responses to pronouns whose gender mismatched the biased referent showed a broad, somewhat bimodal distribution, and the same was true for acceptability judgments of PRO in object-biased items. The second-highest and second-lowest ratings in the reference choice task for each participant are given in Figure 7. The results for the acceptability judgment task are in Figure 8.

Pronouns conflicting with bias again induced wide variability in responses, with most participants inconsistent in their responses. In contrast to Experiment 1, there was also fairly wide variability in responses for PRO in object-biased items. However, all participants still chose the object as the most likely controller for PRO in at least some items with object bias. Similarly, all participants found object control to be acceptable at least some of the time. These results suggest that the main effects in this experiment were similar across individuals. While object control became less likely without established topicality of the



**Figure 8.** *Experiment 2: Acceptability by participant (each dot = one participant)*

object, it was still sometimes preferred and generally acceptable when implicit causality biased towards it.

#### 4. Experiment 3

Experiment 2 demonstrated that object control in temporal adjuncts is still available even when the object’s referent was neither a strongly-established discourse topic nor a logophoric center. Experiment 3 examined whether object control is available even in the absence of supporting context. Previous research has argued that object control interpretations will be largely unavailable or impossible, especially without supporting context (Boeckx, Hornstein & Nunes 2010; Green 2018, 2019a,b; Landau 2021). These claims were made based largely on intuitions of a small number of speakers. This experiment measures whether that intuition holds in a controlled experimental setting.

## 4.1. Methods

### 4.1.1. Participants

Data was collected from 36 participants (23 female, 13 male; mean age = 39.2,  $SD = 12.9$ ).

### 4.1.2. Materials

Materials were modeled after the items in Experiment 2. However, the main clause verbs were modified to remove implicit causality bias in order to make the two characters more equally likely to be mentioned in the adjunct. Although we were not examining the variable of bias, the stimuli still included two versions of each sentence where the characters were reversed. We labeled this manipulation “target character”, but unlike the bias manipulation in Experiments 1 and 2, we expected this to have no effect on the results. An example animate item set is given in (12), and an inanimate set is given in (13).

- (12) (a) **Subject target character:** Sergio saw Alyssa after {sitting/he sat/she sat} in a front-row seat.
- (b) **Object target character:** Alyssa saw Sergio after {sitting/she sat/he sat} in a front-row seat.

### Questions

- (a) Who sat in a front-row seat? (5-point scale from “definitely Sergio” to “definitely Alyssa”)
- (b) If it was {Sergio/Alyssa} who sat in a front row seat, how natural would the above sentence sound, given the context? (5-point scale from “completely unnatural” to “completely natural”)
- (13) (a) **Subject target character:** The crayon was given to Edward after {falling/it fell/he fell} off a chair.
- (b) **Object target character:** Edward was given the crayon after {falling/he fell/it fell} off a chair.

### Questions

- (a) Who or what fell off a chair? (5-point scale from “definitely the crayon” to “definitely Edward”)
- (b) If it was {Edward/the crayon} that fell off a chair, how natural would the above sentence sound, given the context? (5-point scale from “completely unnatural” to “completely natural”)

#### 4.1.3. Procedure and analysis

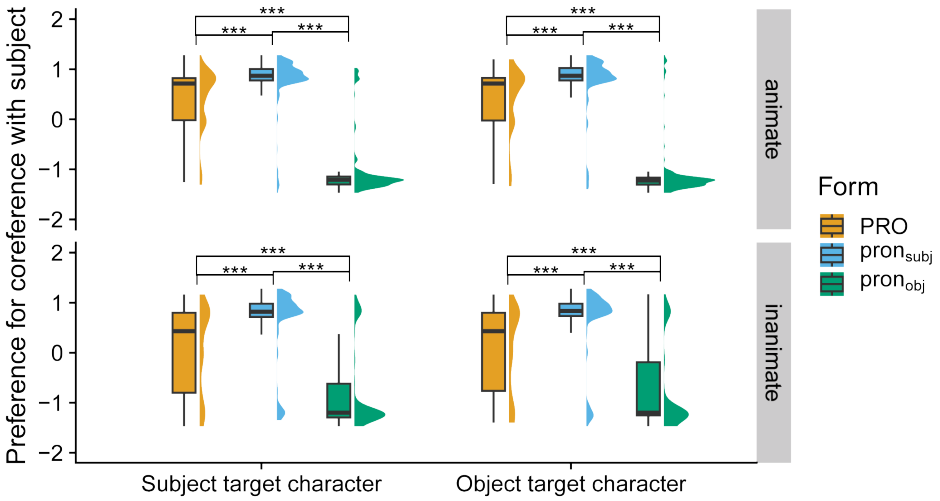
The procedure and analysis for Experiment 3 replicated that of Experiment 2. The final model for reference choice included main effects for form, animacy, and target character, a two-way interaction between form and animacy, and random intercepts for items with random slopes by item for form and target character. The final model for acceptability ratings included main effects for referent, animacy, and form; all two-way interactions and the three-way interaction between the main effects; as well as random intercepts for items and random slopes by item for referent. There was no effect of target character.

## 4.2. Results

### 4.2.1. Reference choice

The average preference for subject coreference in response to the reference resolution question for each condition is illustrated in Figure 9, and the results of the linear mixed-effects model are given in Table 6. We only found an effect of form, with no effects of animacy or target character, and no significant interactions. Both Form1 and Form2 were significant in the model, indicating that PRO was significantly less likely to corefer with the subject than subject-oriented pronouns, but significantly more likely to do so than object-oriented pronouns.





**Figure 9.** Experiment 3: Mean likelihood for critical referential expression to co-refer with the subject of the sentence (z-scored)

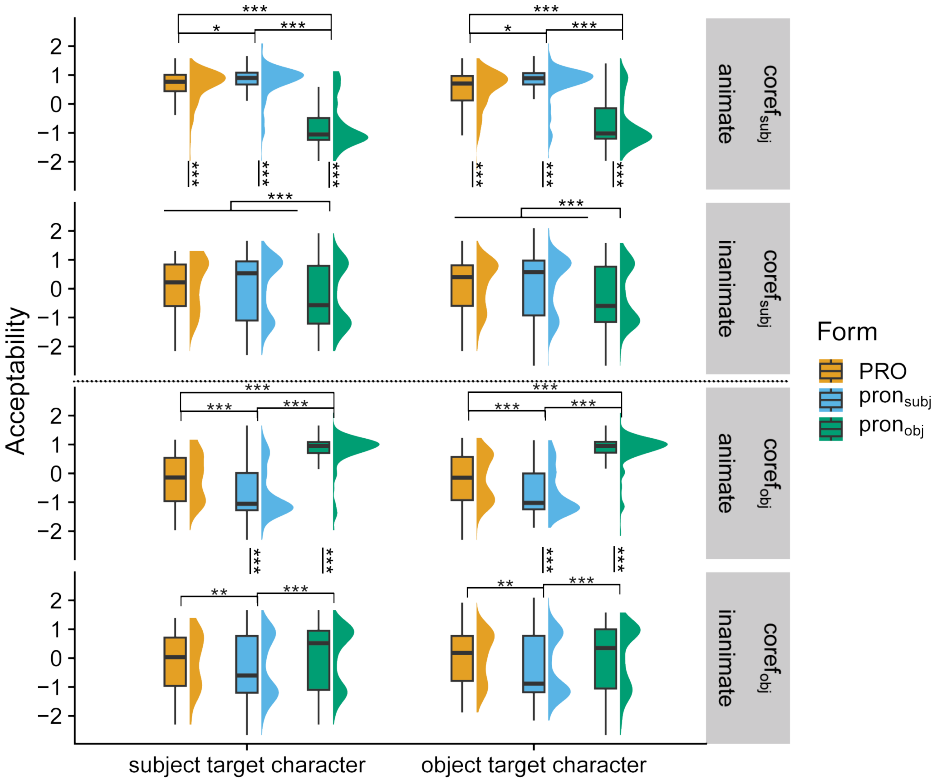
**Table 6.** Experiment 3: Fixed effects for reference choice responses.

| Fixed effect                  | $\beta$ | SE   | df   | $t$   | $p$   |     |
|-------------------------------|---------|------|------|-------|-------|-----|
| (Intercept)                   | -0.00   | 0.02 | 46   | -0.09 | .977  |     |
| Form1                         | 0.63    | 0.07 | 46   | 8.86  | <.001 | *** |
| Form2                         | -0.87   | 0.10 | 46   | -8.43 | <.001 | *** |
| Animacy                       | 0.04    | 0.02 | 46   | 1.53  | .402  |     |
| TargetCharacter               | -0.01   | 0.01 | 1570 | -1.08 | .561  |     |
| Form1:Animacy                 | 0.08    | 0.07 | 46   | 1.18  | .561  |     |
| Form2:Animacy                 | -0.21   | 0.10 | 46   | -2.00 | .205  |     |
| Animacy:TargetCharacter       | -0.01   | 0.01 | 1570 | -0.49 | .799  |     |
| Form1:TargetCharacter         | 0.00    | 0.02 | 1580 | 0.03  | .977  |     |
| Form2:TargetCharacter         | -0.02   | 0.02 | 1579 | -0.93 | .607  |     |
| Form1:Animacy:TargetCharacter | 0.01    | 0.02 | 1580 | 0.43  | .799  |     |
| Form2:Animacy:TargetCharacter | -0.01   | 0.02 | 1579 | -0.48 | .799  |     |

Note: Form1 = pron<sub>subj</sub> vs. PRO; Form2 = pron<sub>obj</sub> vs. PRO.  $p$ -values adjusted for FDR. \* $p$  < .05, \*\* $p$  < .01, \*\*\* $p$  < .001

#### 4.2.2. Acceptability judgments

Mean normalized acceptability ratings are illustrated in Figure 10. The results for the linear mixed-effects model examining acceptability ratings are given in Table 7. We found three-way interactions between form, animacy, and referent.



**Figure 10.** Experiment 3: Mean acceptability ratings (z-scored)

Follow-up comparisons indicated that subject-oriented pronouns (animate:  $\beta = -1.40$ ,  $z.ratio = -17.07$ ,  $p < .001$ ; inanimate:  $\beta = -0.39$ ,  $z.ratio = -4.48$ ,  $p < .001$ ) and PRO (animate:  $\beta = -0.75$ ,  $z.ratio = -9.13$ ,  $p < .001$ ; inanimate:  $\beta = -0.18$ ,  $z.ratio = -2.06$ ,  $p = .039$ ) had higher naturalness ratings for subject coreference than the object coreference. Object-oriented pronouns had higher ratings for object coreference than subject coreference (animate:  $\beta = 1.41$ ,  $z.ratio = 17.15$ ,  $p < .001$ ; inanimate:  $\beta = 0.27$ ,  $z.ratio = 3.14$ ,  $p = .002$ ).

Coreference with the matrix subject received higher ratings for subject-oriented pronouns (animate:  $\beta = 1.42$ ,  $z.ratio = 19.99$ ,  $p < .001$ ; inanimate:  $\beta = 0.34$ ,  $z.ratio = 4.57$ ,  $p < .001$ ) and PRO (animate:  $\beta = -1.24$ ,  $z.ratio = -17.38$ ,  $p < .001$ ; inanimate:  $\beta = -0.39$ ,  $z.ratio = -5.78$ ,  $p < .001$ ) than for object-oriented pronouns. Subject-oriented

**Table 7.** Experiment 3: Fixed effects for acceptability rating responses.

| Fixed effect           | $\beta$ | SE   | df   | $t$    | $p$   |     |
|------------------------|---------|------|------|--------|-------|-----|
| (Intercept)            | -0.00   | 0.02 | 46   | -0.25  | .965  |     |
| Referent               | -0.09   | 0.02 | 46   | -4.11  | <.001 | *** |
| Animacy                | 0.07    | 0.02 | 46   | 4.34   | <.001 | *** |
| Form1                  | -0.04   | 0.02 | 3330 | -2.03  | .064  |     |
| Form2                  | -0.06   | 0.02 | 3330 | -2.80  | .009  | **  |
| Referent:Form1         | -0.36   | 0.02 | 3329 | -17.10 | <.001 | *** |
| Referent:Form2         | 0.51    | 0.02 | 3329 | 23.98  | <.001 | *** |
| Referent:Animacy       | -0.04   | 0.02 | 46   | -1.79  | .106  |     |
| Animacy:Form1          | 0.00    | 0.02 | 3330 | 0.11   | .988  |     |
| Animacy:Form2          | -0.00   | 0.02 | 3330 | -0.02  | .988  |     |
| Referent:Animacy:Form1 | -0.22   | 0.02 | 3329 | -10.23 | <.001 | *** |
| Referent:Animacy:Form2 | 0.32    | 0.02 | 3329 | 15.24  | <.001 | *** |

Note: Form1 = pron<sub>subj</sub> vs. PRO; Form2 = pron<sub>obj</sub> vs. PRO.  $p$ -values adjusted for FDR. \* $p$  < .05, \*\* $p$  < .01, \*\*\* $p$  < .001

pronouns also received higher ratings for subject coreference than PRO did for animate items ( $\beta = 0.19$ ,  $z.ratio = 2.61$ ,  $p = .025$ ), but not for inanimate items ( $\beta = -0.04$ ,  $z.ratio = -0.59$ ,  $p = .827$ ). On the other hand, coreference with the matrix object received higher naturalness ratings for object-oriented pronouns than for subject-oriented pronouns (animate:  $\beta = -1.39$ ,  $z.ratio = -19.45$ ,  $p < .001$ ; inanimate:  $\beta = -0.32$ ,  $z.ratio = -4.20$ ,  $p < .001$ ), as well as for PRO in animate items (animate:  $\beta = 0.92$ ,  $z.ratio = 12.91$ ,  $p < .001$ ; inanimate:  $\beta = 0.06$ ,  $z.ratio = 0.82$ ,  $p = .691$ ). PRO received higher ratings than subject-oriented pronouns with object coreference in both animate and inanimate items (animate:  $\beta = -0.47$ ,  $z.ratio = -6.54$ ,  $p < .001$ ; inanimate:  $\beta = -0.25$ ,  $z.ratio = -3.41$ ,  $p = .002$ ).

Animacy affected all three reference forms. Subject-oriented pronouns received higher ratings in animate than inanimate items with subject coreference ( $\beta = 0.66$ ,  $z.ratio = 7.83$ ,  $p < .001$ ), but higher ratings for inanimate items than animate items with object coreference ( $\beta = -0.36$ ,  $z.ratio = -4.65$ ,  $p < .001$ ). Object-oriented pronouns showed the opposite, with higher ratings for animate than inanimate items with object coreference ( $\beta = 0.71$ ,  $z.ratio = 9.29$ ,  $p < .001$ ) and higher ratings for inanimate items than animate items with subject coreference ( $\beta = -0.42$ ,  $z.ratio = -5.05$ ,  $p < .001$ ). For PRO, subject coreference

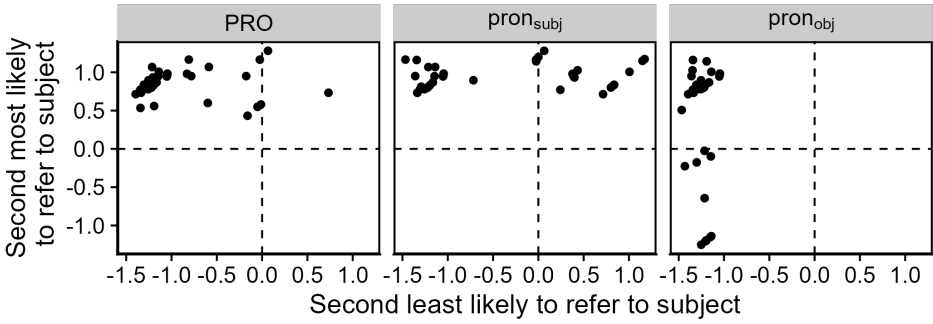
was rated higher in animate than inanimate items ( $\beta = 0.43$ ,  $z.ratio = 5.12$ ,  $p < .001$ ); the effect of animacy with object coreference was non-significant ( $\beta = -0.45$ ,  $z.ratio = -1.91$ ,  $p = .057$ ), but trended toward higher ratings for inanimate items.

### 4.3. Discussion

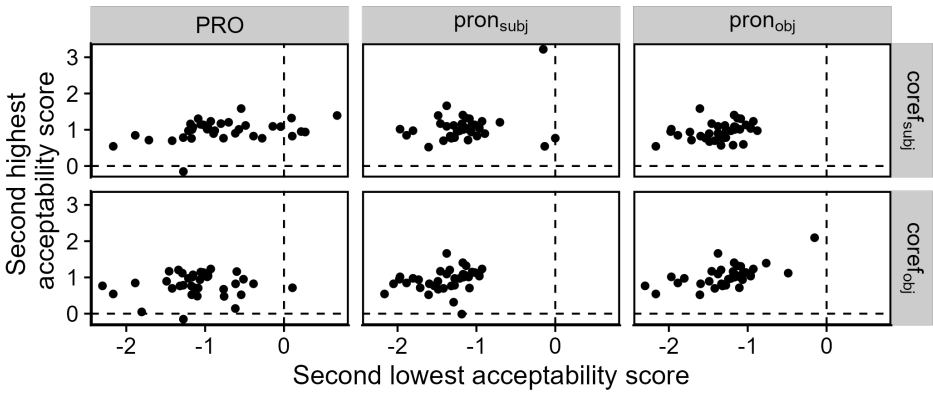
This experiment investigated whether object control is possible for native English speakers even without contextual support. Consistent with previous research (Boeckx, Hornstein & Nunes 2010; Green 2018, 2019a,b; Landau 2021), the results of Experiment 3 indicate that subject control is preferred for temporal adjuncts in a neutral context. However, the reference choice results indicate that preference for coreference with the matrix subject was not as strong for PRO as it was for a pronoun matching the subject's gender. This suggests that object control was at least sometimes still considered a (possibly) weak option for PRO. This is confirmed by the acceptability rating results. Although subject-control interpretations of PRO received higher ratings than object-control interpretations, PRO received higher ratings for coreference with the matrix object than subject-oriented pronouns did. This once again suggests that object control was viewed as possible, despite the object being neither a logophoric center nor a strong topic.

Importantly, the acceptability of object control was not due to any bias toward reference to the object, since we removed implicit causality from our items. The lack of effect of target character on the results indicates that bias was successfully removed.

The effect of animacy in this experiment was similar to the effect found in Experiment 2. For overt pronouns, participants allowed for inanimate objects to be referred to with gendered pronouns more often than for people to be referred to with non-gender-matching pronouns. For PRO, subject control was again more acceptable with animate controllers, but animacy of the object did not improve object control.



**Figure 11.** Experiment 3: Subject preferences by participant (each dot = one participant)



**Figure 12.** Experiment 3: Acceptability by participant (each dot = one participant)

#### 4.3.1. Exploratory analyses

Reference choices did not show a bimodal distribution, but acceptability ratings did. Figures 11 and 12 show the second-highest and second-lowest ratings for each participant for reference choice and acceptability, respectively. Because there was no effect of target character, these plots collapse across that condition.

Although some participants consistently chose the character matching in gender as the pronouns' referent, others did so only some of the time. This may have been due to perceived ambiguity in the names' genders or, as discussed above, using gendered pronouns to refer to inanimate objects. All participants exhibited variability in how acceptable it was to use either pronoun, regardless of referent.

More importantly, the results for PRO indicate that almost all participants sometimes selected the object as PRO's referent at least as a possibility, and that object control was found to be acceptable in some items.

## 5. General discussion

The goals of this study were as follows: (i) to confirm the findings of [Janke & Bailey \(2017\)](#) that object control of temporal adjuncts is possible when the object's referent is established as a strong discourse topic; and (ii) to investigate whether topicality and animacy are relevant to controller choice, as predicted by Landau's (2021) Controller Worthiness Scale. This section will discuss each of these questions in turn. It will then discuss the results of an exploratory analysis that casts doubt on Janke & Bailey's (2017) structural-ambiguity account of object control in temporal adjuncts before discussing how the results could be accommodated in other recent theories of adjunct control.

### 5.1. *Object control is available*

The results of Experiment 1 clearly confirm the results of [Janke & Bailey \(2017\)](#). We found that object control in temporal adjuncts is largely available when the referent of the object is established as a strong discourse topic. In fact, object control appears to be even more preferred for object-biased items in our study than in Janke & Bailey's. In Experiment 1 of the current study, nearly all responses showed a strong preference for object control over subject control with a supporting context; Janke & Bailey's results only exhibited 51% object-control choices. This is likely due to the increased bias included in our experiment through both topicality and implicit causality. Importantly, whereas Janke & Bailey's results included only forced binary reference choice responses, our data confirms the availability (and preferability) of object control given a supporting context with both Likert-scale reference choice data and acceptability ratings. Not only was the object the most likely referent in object-biased items, but object control was also judged as highly natural.

The results of Experiments 2 and 3 extend these findings by illustrating that strong topicality is not a necessary condition for object control interpretations. Experiment 2 demonstrated that implicit causality alone is enough to facilitate object control, although to a lesser extent than when implicit causality and topicality are combined. Even in Experiment 3, in which referential bias was removed, object control was sometimes accepted.

These results also experimentally confirm recent work based largely on speaker intuitions (e.g. Landau 2015, 2017, 2021; Green 2019a,b) arguing that temporal adjuncts do not require strict OC by the subject—as was claimed in earlier work (Clark 1990, Hornstein 1999, Pires 2007)—and that non-subject control does not require subject control to be blocked (contra Manzini 1983, Boeckx & Hornstein 2007, McFadden & Sundaresan 2016). Although the embedded predicate in some of our items in the inanimate conditions of Experiments 1 and 2 was semantically anomalous with an animate subject (e.g., *melt* in the example stimulus in (11)), others were neutral, allowing both animate and inanimate subjects (e.g., *teach*). Exploratory analyses (see OSF supplementary files) found no effect of this animacy restriction in Experiment 1. Having a predicate semantically restricted to inanimate subjects did increase the availability of object control by an inanimate object in Experiment 2, but object control was still widely available in items without such an animacy restriction. It was also equally available in animate items where both the subject and object were semantically plausible as controllers. These results together provide evidence that although subject control may be preferred in general, object control is a viable grammatical option, at least with sufficient contextual support.

## 5.2. *The effects of animacy and topicality*

Our second research question was whether animacy and topicality affect the availability of object control in temporal adjuncts. Some previous research argued that NOC is limited to logophoric (human) antecedents (Williams 1992; Landau 2015, 2017; Green 2018, 2019a,b), while other research argued that NOC can involve inanimate controllers

(Donaldson 2021a,b). Landau's (2021) Controller Worthiness Scale explicitly claims that animate subjects should be the most likely controllers for PRO, allowing predicative, logophoric, and topic control. Inanimate subjects are also possible predicative controllers and weak topic controllers, but are unavailable for logophoric control. As for non-subject controllers, Landau claims that human antecedents should be available as logophoric controllers or weak topic controllers. However, inanimate non-subjects should be far less likely than animate non-subjects to control PRO, due to their inability to be logophoric perspective-holders and reduced ability to be topics.

We found an effect of animacy for subject control in Experiments 2 and 3, in line with previous findings (Kawasaki 1993, Landau 2001, Parker, Lago & Phillips 2015). Animate (human) subjects were more acceptable as controllers than inanimate subjects. This may give some support for Landau's Controller Worthiness Scale. Subject control is more available when both predicative and logophoric control are possible (as with the animate subjects in Experiments 2 and 3) or when both predicative and topic control are possible (animate and inanimate subjects, Experiment 1), compared to when only predicative control is (inanimate subjects, Experiments 2 and 3). But the possibility of all three types of control (animate subjects, Experiment 1) does not seem to improve acceptability over just two types (inanimate subjects, Experiment 1), since there was no effect of animacy in Experiment 1.

However, the Controller Worthiness Scale does not fully predict the results seen for object control. Having the object as a strong topic did increase the availability of object control—object control appeared to be more available in Experiment 1 than in Experiment 2. However, although Landau (2021) acknowledges that logophoricity and topicality are not both necessary for NOC (including object control), he does assume that having at least one is (see also Landau 2025). This was not the case in our experiments. In Experiment 2, implicit causality biasing towards reference to the character named by the object was sufficient to strongly increase object control interpretations. Even in Experiment 3, which lacked any bias towards coreference with the object, object control was still sometimes available. In addition, the animacy of the object had no effect on object control interpretations across



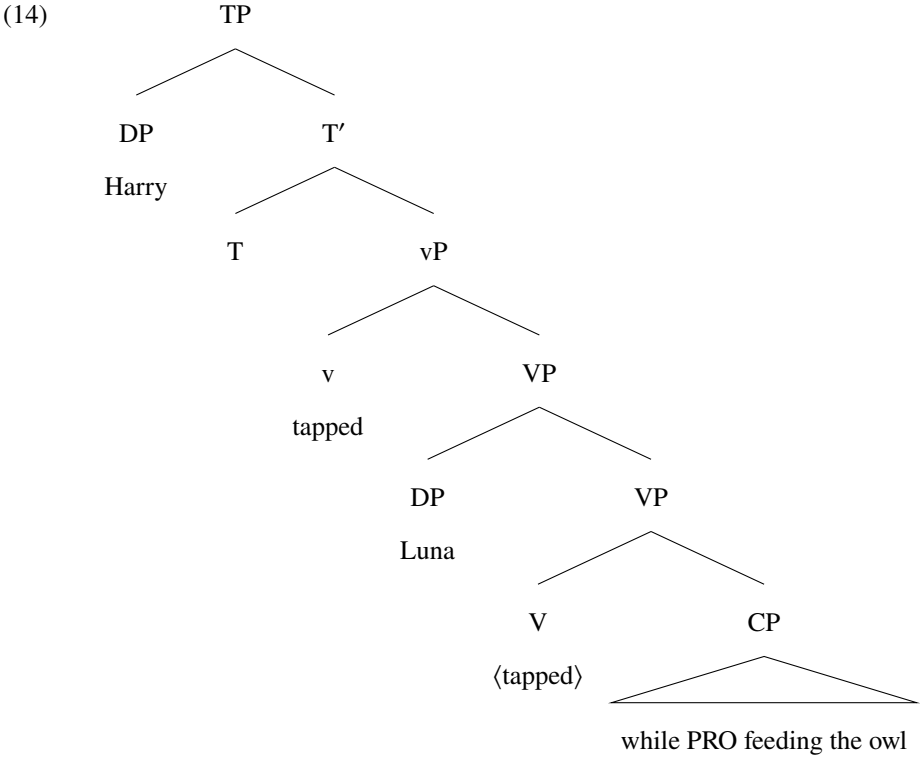
any of the experiments, even though the Controller Worthiness Scale predicts that animate antecedents should be more available for NOC than inanimate ones.

One potential reason that animacy did not have an effect on object control, especially in Experiments 2 and 3, was that despite the object being animate, it was nevertheless not linguistically established as a perspective holder, thus limiting the availability of logophoric NOC. In addition, the objects were not established as strong topics in Experiments 2 and 3, limiting the availability of topic NOC by the object. Because of that, even if animate referents do in fact make better logophoric centers and topics than inanimate ones, that may have been irrelevant in Experiments 2 and 3. This does not necessarily explain why animacy did not have an effect in Experiment 1, however. In Experiment 1's animate items, the preceding context not only established the object's referent as a topic, but also as a perspective holder, increasing the possibility of logophoric control. That was not the case in inanimate items. Nevertheless, object control in inanimate items was just as available and acceptable as in animate items. This indicates that with the object established as a strong topic, the availability of logophoric control in animate items does not increase the availability of object control. Therefore, although topicality is sufficient for licensing object control, neither it nor logophoricity is necessary.

In sum, the Controller Worthiness Scale does capture important generalizations about NOC in adjuncts. Object control in temporal adjuncts, as a specific type of NOC, is more likely to be available with logophoric or strongly topical antecedents. But the Controller Worthiness Scale is too strong, since object control is possible even in the absence of logophoricity and topicality.

### ***5.3. The availability of object control is not due to a structural ambiguity***

Janke & Bailey (2017) argued that object control in temporal adjuncts is not an instance of NOC, but is instead available due to a structural ambiguity, with object control resulting from lower attachment of the adjunct, below the position of the object, as illustrated in (14), repeated from (6).



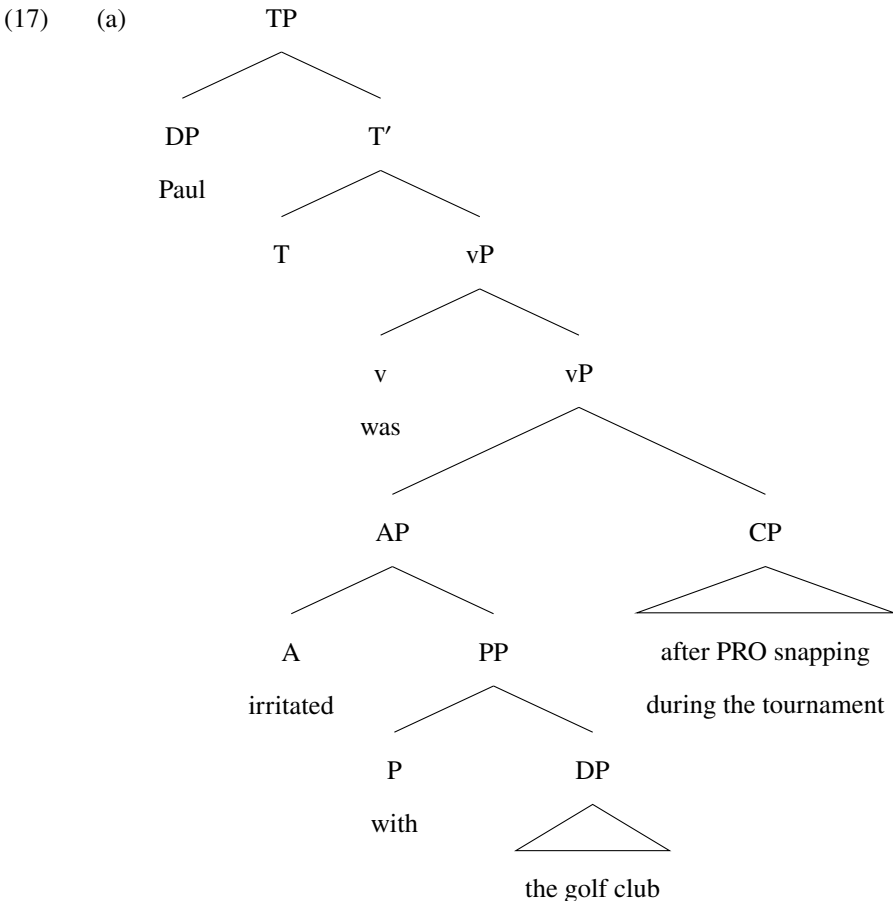
This structural analysis could at least partially explain why we found such a limited effect of animacy and topicality in our results; object control in a structure such as (14) would be OC—predicative, in Landau’s (2015, 2021) terms—rather than NOC, and would therefore not require a logophoric or topical antecedent to PRO. However, in addition to the issues with this account discussed by Landau (2021), the structure in (14) would not even be possible with some of the items in the current experiment.

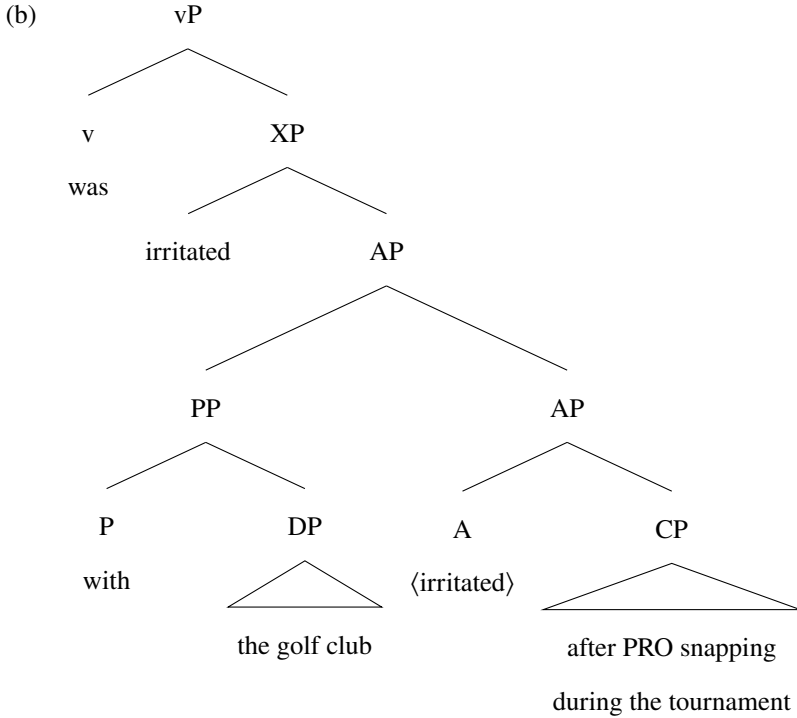
Although some of our object-biased items such as those in (15) biased towards control by a direct object, as in Janke & Bailey’s (2017) experiment, we have been using the term “object control” loosely to refer to any type of object, whether a direct object, indirect object, or object of a preposition, as in (16). This was especially true for inanimate items.

- (15) (a) Alyssa noticed Sergio after sitting in a front-row seat.  
 (b) Anita forgave Jeff after butchering her haircut.

- (16) (a) Shirley was happy with Todd after bringing a salmon home from the Pacific.  
 (b) Edward was disappointed with the crayon after melting in the heat.  
 (c) Paul was irritated with the golf club after snapping during the tournament.

Crucially, the relevant hypothesized structural ambiguity that would allow the direct object to c-command and therefore syntactically control PRO in (14) would not allow for syntactic object control by a prepositional object. This is illustrated for (16c) in (17). In (17a), because *was* occupies the voice position, it is unlikely that the adjunct could occupy a position lower than *with the golf club*. Even if it could, perhaps due to a similar structure to the one in (14) and illustrated in (17b), *the golf club* would still be embedded inside a PP and would therefore not be in a position to c-command and syntactically control PRO.





The structural ambiguity account put forward by [Janke & Bailey \(2017\)](#) would therefore predict object control to be more available in items with a direct-object controller than in items with a prepositional object controller. We ran an exploratory analysis to test this prediction and found no significant difference between the two object types in terms of reference choice or acceptability judgments, regardless of contextual bias (see OSF for full results). This provides some evidence against a structural ambiguity account of object control, although a more systematic comparison of object types could reveal additional relevant findings.

#### ***5.4. Potential accounts of object control in temporal adjuncts***

As already discussed, the results from the current experiment present challenges for the Controller Worthiness Scale as an explanation for when NOC and other cases of object control will be available in temporal adjuncts. However, these results are not incompatible

with Landau's (2021) theory of adjunct control overall. Indeed, the Controller Worthiness Scale was not intended as a theory of what is available grammatically, but as a processing theory for finding the referent of PRO in NOC adjuncts. The contribution of the current experiments, therefore, is to demonstrate that NOC is also possible with other forms of contextual support than topicality and logophoricity.

#### 5.4.1. NOC PRO is *pro*

What is the nature, then, of PRO in NOC structures such as object control in temporal adjuncts? Landau (2021) discusses previous examples of object control by inanimate objects into temporal adjuncts. Although he is skeptical that they are, in fact, acceptable, he gives as a possible hypothesis (citing Herbeck 2019, Paz 2019) that NOC PRO in adjuncts is really just the null pronoun *pro* (see also Herbeck 2015, 2021; McFadden & Sundaresan 2018). If NOC PRO is really just a null pronoun, then it should be subject to similar referential restrictions as other minimal pronouns. This would naturally explain why implicit causality alone is enough to license object control interpretations. Even if the object's referent is not an established topic or logophoric perspective holder, if that referent is the most likely to be mentioned, then readers will interpret the null pronoun as coreferring with the object, regardless of its referent's animacy (Vender, Nardon & Venagli 2025).

The hypothesis that NOC PRO is just a null pronoun is still compatible with an overall bias for subject control in absence of contextual support. Not only is syntactic (predicative) control by the subject an available and perhaps preferred option, but pronouns also have been shown to have a subject bias (see, e.g., Lam & Hwang 2024). It is also compatible with the ideas of the Controller Worthiness Scale, since minimal pronouns have been shown to favor topical antecedents (Frascarelli 2007, Ng, Hwang & Cui 2025)

However, as Landau (2021) points out, this account still leaves the question of why null subjects more strongly favor human antecedents in NOC contexts than in other contexts. One potential answer could be that there is something else in the NOC structure that favors

logophoric interpretations (cf. [Koopman & Sportiche 1989](#)). This could be a position for the logophoric antecedent in the periphery, as has been proposed for so-called exempt anaphors ([Charnavel 2020](#), [Baker & Ikawa 2024](#)). However, it is unclear why that position would be more likely to be present (but still not required) for NOC structures than in other structures containing *pro*. These questions are left for future work. The present results simply demonstrate that object control in temporal adjuncts—and by hypothesis NOC in general—is available whenever an NOC antecedent is highly likely, whether this is due to topicality, logophoricity, or other contextual factors.

#### 5.4.2. Movement-based accounts

The results of the current study are compatible with Movement-based accounts of adjunct control ([Boeckx, Hornstein & Nunes 2010](#), [Green 2019a](#)), or the more recent Form-Copy theory ([Chomsky 2021](#), [Chomsky et al. 2023](#)), which also argue that PRO in NOC is the null pronoun *pro*. In these accounts, *pro* in NOC is expected to behave similarly to other null pronouns, so it would be unsurprising that contextual constraints other than topicality and logophoricity would affect NOC interpretations. It would therefore follow that such contextual constraints would make object control more likely. However, for extensive discussion of challenges these accounts face, see [Landau \(2021, 2024\)](#).

#### 5.4.3. Object control is not a grammatical illusion

A final possible explanation for the acceptance of (inanimate) object control in the current study is that it reflects a grammatical illusion, rather than a genuine option made available by the grammar (for discussion on the debate between grammatical versus processing accounts of other phenomena such as ellipsis mismatches and agreement attraction, see [Phillips 2013](#), [Phillips & Parker 2014](#)). This would be consistent with accounts stating that based on the position of temporal adjuncts in the structure, only subject control should be available ([Boeckx, Hornstein & Nunes 2010](#), [Fischer & Høyem 2022](#)). [Gerard \(2021\)](#) found that children are more likely to allow object-control when the task used to measure

their interpretations has high memory demands; when the demand is lessened, they have more adult-like (subject-control) interpretations. Gerard argues that the grammar of these children (and of adults) only allows subject control, and that their acceptance of object control is due to an immature parsing mechanism, influenced by working memory capacity. Similarly, it is possible that in the current experiment, contextual bias outweighed grammatical constraints in a way that led people to accept object control extra-grammatically. This could be true just for object control by an inanimate object or for both animate and inanimate items. In either case, if acceptance of object control is due to a grammatical illusion, then the current results would not present a genuine challenge to Landau's (2021) theory or his Controller Worthiness Scale.

Previous research has argued that grammatical but rare constructions can be subject to within-experiment adaptation effects, while truly ungrammatical structures are not (Sprouse 2007, Chaves & Francis 2024, Lu, Frank & Degen 2024). Therefore, in the current experiment, if acceptance of object control (whether animate or inanimate) were due to a grammatical illusion, we would expect lower ratings compared to (animate) subject control that did not change throughout the experiment. If, however, object control is rare but grammatical, we would expect ratings for object control to rise throughout the course of the experiment. To test this possibility, we ran exploratory analyses to measure the effect of trial order (coded as a continuous, linear variable, or as a log-transformed variable) on reference choice and acceptability ratings in the PRO items (see OSF for full results). In Experiment 1, we found a significant interaction between linear order, referent, bias, and animacy on acceptability ratings ( $\beta = -0.06$ ,  $SE = 0.02$ ,  $t = -3.16$ ,  $p = .002$ ); as the experiment progressed, object control of PRO became more acceptable and subject control less acceptable in inanimate items with object bias. Similarly, subject control became more acceptable in inanimate items with subject bias. Similar results were seen with the log-transformed trial order. In Experiment 2, there was a main effect of log-transformed trial order on reference choices ( $\beta = -0.06$ ,  $SE = 0.03$ ,  $t = -2.52$ ,  $p = .001$ ); PRO was rated as less likely to corefer with the subject later in the experiment (with no interaction with bias

or animacy). Both linear and log-transformed order had positive effects on acceptability of object control with object bias (linear effect:  $\beta = 0.13$ ,  $SE = 0.04$ ,  $t = 3.14$ ,  $p < .01$ ), and on acceptability of subject control with subject bias ( $\beta = 0.09$ ,  $SE = 0.04$ ,  $t = 2.17$ ,  $p = .03$ ). Trial order did not affect acceptability of items where the referent conflicted with bias. Experiment 3 showed no effects of trial order.

These exploratory results demonstrate adaptation or satiation effects for object control: in Experiments 1 and 2 where contextual bias was present, object control became more acceptable over time within the experiment. This suggests that object control is a grammatical option; it may have exhibited low acceptability in previous studies just because it is rare in the input. The fact that this adaptation effect was only visible for inanimate items in Experiment 1 may be because of the availability of logophoric and topic control in animate items, which may have increased the acceptability of object control in these items earlier in the experiment. In Experiment 2, when topicality and logophoricity of the object was lessened, both animate and inanimate items exhibited adaptation. The lack of similar effects in Experiment 3 is unsurprising, since there was never any bias toward object control in the items that would lead to adaptation. The presence of adaptation effects in the first two experiments suggests that object control by both animate and inanimate antecedents is a grammatical option in temporal adjuncts.

### 5.5. *Future work*

This research has demonstrated that object control is available and acceptable in temporal adjuncts given a sufficient supporting context, and that animacy of the antecedent has no effect on PRO's interpretation. It was assumed that object control is an instance of NOC, since the object is not in a position to c-command PRO in the adjunct. If this is the case, then we would expect similar results for extra-sentential NOC controllers. However, Janke (2018) found that although participants could be led to accept object control by making the object's referent highly topical, they did not accept extra-sentential controllers, even when strongly cued toward that interpretation. Still, Green (2019b) discussed several examples



of sentence-external NOC antecedents to PRO in temporal adjuncts, arguing that it is a grammatical option. These extra-sentential NOC controllers were always logophoric, however. Landau (2021) discusses some examples of extra-sentential control by inanimate objects, but questions their grammaticality. Future research could more systematically probe the effect of contextual bias on extra-sentential NOC to confirm whether it and object control in temporal adjuncts are both, in fact, instances of the same type of NOC.

Another area where future research is needed is the role of individual differences in the acceptability of object control in adjuncts. Gerard & McDaniel (2025) found that lesser working memory capacity in children is associated with more frequent object control interpretations. Although we found that object control with a supporting context was broadly available across participants, there was some individual variation. We did not collect detailed background information on individual participants and were therefore limited in our ability to explore whether individual differences affected reference choice or acceptability ratings. However, we did run exploratory analyses to examine the effects of the demographic information automatically provided by Prolific: age, gender, and ethnicity (see OSF). We found that object control was rated as more acceptable by older participants compared to younger participants in Experiment 3. There were no other clear effects on the interpretation of object control. Other effects applied to rating of PRO overall or were limited to interpretations of pronouns conflicting with contextual bias. It is unclear why age would have an effect on these interpretations. Other individual differences beyond the few factors explored here may also lead to greater acceptance of object control. We leave these questions to future research.

Finally, although we found that object control is highly available and acceptable in offline judgments, given a supporting context, object control may still be more difficult to process than subject control. In addition, animacy may affect processing in line with the Controller Worthiness Scale in a way that is not apparent when examining offline judgments alone (for a similar distinction between online and offline processing of pronouns in Italian, see Vender, Nardon & Venagli 2025). The study of object control in adjuncts would therefore

benefit from future research on its online processing. This may also shed light on processing accounts of non-subject control in adjuncts, such as that given by Donaldson (2021a,b).

## 6. Conclusion

This study confirmed and extended previous findings (Janke & Bailey 2017) that demonstrated that object control in temporal adjuncts is available and acceptable, given a supporting context. Furthermore, the study found that topicality and logophoricity of the object are not necessary conditions (either jointly or separately) for object control to occur, as suggested by Landau's (2021) Controller Worthiness Scale. Object control was still largely available and acceptable when supported by implicit causality alone. The results are not explainable by a structural ambiguity account such as that of Janke & Bailey (2017). Instead, they are consistent with accounts positing that NOC PRO, as in object control of temporal adjuncts, is simply a null pronoun *pro* whose reference is determined by several competing constraints (Herbeck 2015, 2019, 2021, Paz 2019).

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**Data availability statement.** Full materials, data, and analysis scripts for this study can be found on OSF: [https://osf.io/wzv3a/?view\\_only=49e37b6b25444c7f9a63dd7b65deed92](https://osf.io/wzv3a/?view_only=49e37b6b25444c7f9a63dd7b65deed92).

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